

# ABSTRACT

Tremendous advancements made in Artificial intelligence (AI) and Machine Learning (ML) applications results in high requirement of massive and accurately annotated datasets for training models as the performance is highly dependent on the quality of training data for developing future dependable AI models. However, managing a large scale annotation project has always been challenging due to the difficulties in distributing work to a large group of annotators in a manual fashion, lack of unified quality control and fragmented workflow with minimal traceability on projects and complicated payment system.

To alleviate these problems, a cloud based data annotation management system (the Annotix) supporting end-to-end annotation workflow automation is presented. This system allows users to upload image, video, audio and text datasets, check upload data integrity, auto-create annotation tasks with user adjustable project configurations, and distributes them among annotators through a well-structured workflow, saving labor force and accelerating projects.

Annotix uses a three-tier QA process, L1 Annotators, L2 reviewers, L3 Senior reviewers. Annotators will carry out the first labelling pass. Review will then wait L2 Validity annotation Image. The bulk of the labelling will have L2 review. Deliver back for corrections, suggestions, use of reviewer names.

Apart from that the platform can provide organization/project management, dataset management, automatic task creation, task assignment, calculation of payments, reports and graphical dashboards. Role Based Access Control (RBAC) and JWT authentication mechanisms has been implemented in such a way that roles can be handled flexibly. Audit logs must log up/download, distribution, annotation submitting, the verdict after reviewing, payment issues and admin works. React.js, Spring Boot and PostgreSQL has been used as a common DB for collection of modules that are loosely coupled by the secured REST API.

Distributed collaboration: can be performed by distributed annotators and reviewers in the virtual disaster annotation cloud. The dynamically updated dashboards allow project administrators to quickly fetch real-time information concerning the progress, quality and rate of the annotation workforce, payment and statistical activities, while the system automates payment transactions.

To conclude, Annotix is a flexible and scalable annotation management platform that combines dataset management, workflow automation, quality management, workforce management and project analytics in a single platform. The Annotix platform transforms scattered manual processes into a coherent intelligent platform to achieve the high quality and efficiency of annotation.

Keywords: Artificial Intelligence, Machine Learning, Data Annotation, Annotation Management System, Cloud Computing, Workflow Automation, Quality Assurance, Spring Boot, React.js, PostgreSQL, REST API, JWT Authentication.

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## CHAPTER 1

### INTRODUCTION

#### 1.1 Introduction to the Project

Intelligent system, for example image and speech recognition, natural language processing, autonomous driving, medical diagnosis and predictive analytics are possible by rapidly growing emerging technology of AI and ML. They built AI models which are totally based on the need of large voluminous of structured data with correct label. Data annotation as an integral part of supervised machine learning including labelling of raw data such as image, video, text and audio, etc. provides the data sets for architecture training. These annotated datasets allow AI model to recognize not only pattern but learn the relationships and reach the kind of statement that can only be proven by analogy.

With the proliferation of AI use cases, companies and organizations are now requiring much larger amount of annotated data. Tens or hundreds of thousands of data points even millions of data points are difficult to manage using the traditional annotation processes and workflows. Generally, the entire annotation process through traditional data annotation workflows include use of spreadsheets, shared folders, email communication and manual task assignment. These all can lead to suboptimal task assignment and a significant amount of duplicated work, inconsistent annotation, limited collaboration opportunity, late quality review, and no visibility into progress. Due to less visibility, it is very hard to manually track productivity, review status, calculate pay to the workers, thus losing operational efficiency.

To circumvent the above complications, an online software for so called data annotation cloud tools have been developed recently to arrange annotation tasks. They provide a single platform for all stages of an annotation project enabling supervisors to upload data sets, set up an automated job dispatch, observe ongoing work, simulate multi-pass round-trip quality review routines, and administrate users in a freely accessible web interface.

Annotix is a cloud-based end-to-end data annotation management platform to manage complete annotation lifecycle, including project setup, annotation task creation, team annotation process, and review management. It offers a seamless integrated platform to set up annotation projects, upload dataset, configure the task allocation strategies, and manage a team of annotation specialists for a given project. Given an uploaded dataset, the platform automatically splits it into smaller annotation tasks and distributes to each available annotator.

There is also an established multi-stage review system in this platform. When an annotator finished a task, they need to send it to some annotators to review. The reviewing annotator can agree, dis agree, or ask for change of current annotation on this task, and track those changes. This process will improve the quality of annotations.

Furthermore, Annotix provides full project monitoring and workforce management. It offers a variety of visualization dashboards over project progress, annotation statistics, worker performance, task completion level, productivity, etc. And calculates annotator's payment

automatically according to annotation task number without manual payment calculation, which makes the workforce management much easier and transparent for project managers. Role-based access control also be adopted to protect the annotation process.

The system has been built with contemporary web technologies to favour scalability, security and maintainability. A Spring Boot powered backend application server implements the business logic, user authentication, project management, task annotation, review process, REST interface and API business; whereas a React powered frontend implementation provides an intuitive and responsive user interface and client interaction. The information is stored in a relational database, PostgreSQL shall be used as project management information, user information, annotation recording, and system metadata.

In summary, Implementing Annotix into a data annotation project allows organizations to effectively manage large quantities of data annotation tasks, automate annotation task assignment, facilitate enhanced collaboration among human annotators, streamline payment processing, implement quality assurance measures, and produce project management analytics, all on one easy to use platform. Replacing multiple manual steps needed to manage an annotation project, Annotix centralizes and automates the entire annotation process on one cloud based platform. This allows organizations to deploy and build high-quality annotation datasets more effectively so they can create dependable AI/ML models.

## 1.2 Statement of the Problem

AI and ML progress in recent years is very rapid; consequently, the requirement for huge number of annotated data set of good quality is much higher. To develop an AI based solution, it requires huge number of unambiguous and well annotated data sample to build good and fast learning algorithms. Managing of entire annotation process (i.e. Annotation lifecycle) is a challenge for full fledged automation because of lacking of central platform where all activities like project management, assignment planning, quality check process, manpower management may take place.

Similarly, majority of the current annotation tools are only designed to provide annotation interfaces and not all the other factors involved in the annotation process. Activities like managing the dataset, assigning labelling tasks to annotators, multi-tier of quality reviews, payment calculation, and project tracking.

Current annotation procedures depend on human resources. It requires a project owner to partition the data set into annotation tasks, assign those tasks to annotators, monitor the progress, evaluate annotation quality, compute workers' payment, and so on. It is more and more time-consuming and impossible to coordinate project efforts to annotate a large data set. Also, the annotation distribution among works would probably be unequal so that the whole annotation process may last for a very long time (as it takes very long to complete project annotation). Increasing project size makes labor-consuming annotation process extremely difficult.

An additional drawback is the lack of an end-to-end quality check in the present systems. The currently available annotation environment offers only a single review procedure, which makes errors difficult to identify and is hard to sustain quality when annotating large dataset collections. And low quality annotation may appear in the final dataset and further impair the learning model performance.

Another problem we anticipate is scalability. Without a systems that supports automated splitting of tasks, intelligent task assignment, task locking, and centralized monitoring, we do not believe we could support a large annotation teams (thousands of annotations). Administrators would spend a lot of time in the coordinating, and annotators would spend a lot of time waiting, or inadvertently doing duplicated work.

Security/trust/and accountability, along with the other issues, are equally pressing. Many of the current solutions do not track annotation history, the history of which user completed which task, how the annotations are being reviewed, and how much the annotators are being paid. Without this information, it is impossible to (or at least very difficult to) show that the work has been done, to settle disputes, to assess the performance of the annotators, or to pay more for better annotation. It is also very difficult to specify a role-based access control system and therefore a more secure system, as the data in some projects can be very sensitive.

To overcome the above problems, Annotix was built as a host-cloud based end-to-end data annotation management platform that automates the entire annotation lifecycle. This platform enables centralized dataset management, automatic task creation/assignment, logical multi-tier (L1/L2/L3) reviews, quality checks, role-based access, payment monitoring, audit logs, and reporting of project status all in one integrated platform. This all-in-one platform empowers organizations to automate, streamline and scale up data annotation projects, enhance annotation accuracy and make the entire annotation process transparent and efficient.

### 1.3 System Specifications

Building Annotix on a suitable hardware and software platform is essential to delivering a scalable, secure and high-performing system. The platform follows a modern client-server architecture built on established web technologies. The hardware and software environments recommended for development and deployment are described below.

#### 1.3.1 Hardware Specifications

##### Development Environment

- Processor: Intel Core i5 (10th generation or above) / AMD Ryzen 5 or higher
- RAM: 8 GB minimum (16 GB recommended)
- Storage: 256 GB SSD (512 GB SSD recommended)
- Graphics: Integrated graphics (a dedicated GPU is optional and only relevant for future AI-assisted features)
- Display: 1920 × 1080 Full HD resolution
- Network: Broadband internet connection, minimum 20 Mbps

##### Server Environment

- Processor: Quad-core or higher
- RAM: 16 GB minimum
- Storage: 512 GB SSD or equivalent cloud storage
- Operating System: Ubuntu Server 22.04 LTS / Windows Server 2022
- Database Server: PostgreSQL
- Network: High-speed connection with HTTPS support

### 1.3.2 Software Specifications

| Component        | Specification                      |
|------------------|------------------------------------|
| Operating System | Windows 11 / Ubuntu 22.04 LTS      |
| Frontend         | React.js, TypeScript, Tailwind CSS |
| Backend          | Java, Spring Boot                  |
| Database         | PostgreSQL                         |
| Build Tool       | Maven                              |
| API Architecture | RESTful APIs                       |
| Authentication   | JWT (JSON Web Token)               |
| Version Control  | Git & GitHub                       |
| IDE              | Visual Studio Code, IntelliJ IDEA  |
| API Testing      | Postman                            |

### System Requirements

Annotix is a cloud based application with a web interface. The React.js user interface interacts with Spring Boot back end via secure REST interfaces containing the application's business logic and processing for tasks, reviews and payments, and reads and writes data a single centralised PostgreSQL database. It is built for a multiuser environment.

It is role-based and thus applicable for Company Clients, L1 Annotators, L2 Reviewers, L3 Super Reviewers and System Admins. Secure authentication, encryption enabled communication and a centralised storage of data on the server make the system nonpublic, reliable and concrete; offering a stable platform for further development, deployment, maintenance and extension of the Annotix platform.

## CHAPTER 2

### LITERATURE SURVEY

An extensive study of the current state of art was carried out to document the technologies and approaches that exist for AI / ML data annotation management systems. In particular, the area of workflow automation for annotation, task management, quality management, collaborative annotation, cloud based deployments and project management was investigated. These results from the survey have then been used to model the architecture of existing annotation platforms and to design the Annotix application.

The boom development of AI and machine learning has increased the requirement for more accurately labelled data. Supervised learning models can only be properly and effectively trained on correct labelled data. However, it is still a difficult task to organize hundreds of thousands or even millions data samples for annotation, and to ensure their completion on time. Researchers and commercial sites put forward a number of ways to improve the annotation process.<sup>3,4</sup>

The annotation platform, developed by Sapien AI (2023), was a cloud based solution that is designed for the enterprise annotation task and that offers features such as automatic task distribution, centralized project management and a defined QA process in order to allow for scaled high throughput annotation projects in a uniform way. The experiment found that automated task assignment largely reduces the amount of manual administration needed in large-scale annotation tasks and was a more efficient use of resources when applied to multiple annotators.

Similarly, in Digital Bricks (2023) it stated that the adoption of AI technology needs access to the BLANK label data of a higher quality with high band width. It was indicated that human intervention in the loop view process was necessary to achieve the quality of the annotation; finally, the annotation management workflow could better automated than annotation itself.

In Annotate AI (2023), a general web-based framework for annotation has been designed which includes management of worker, assignment of worker and management of project. During testing, the automatic assignment of annotator has been found. As compared to who does not, an increased annotation rate is anticipated from automatic assignment of workers as it would lead to more effective handling of the annotation task and efficiency would be gained by reduced working time while worker is idle. In case of workflow management, live monitoring has been found to be critical.

Earlier study carried out by Sapien AI (2022) worked on the crowdsourcing multiple annotators for large scale annotation of AI training dataset where the same data was annotated by different annotators in an overlapped manner thus reducing the time for annotation of the entire dataset. It also developed approaches for annotation agreement across multiple annotators for annotation consensus and to distinguish ambiguous data for the second/third annotation review.

Besides open source annotation tools, many commercially available annotation platforms contribute a lot to managing annotations like Label Studio, CVAT, SuperAnnotate, Scale AI, Labelbox etc. Typically these platforms offer specialized tools for each type of annotation (image, video, audio or text), and allow collaboration between users working on it. While cloud deployment, collaborative annotation, annotation versioning, and export options for data are widely available in annotation tools nowadays, managing projects, making payments, workforce management,

multiple levels of reviews and other infrastructure aspects are frequently dealt with using other tools.

In terms of annotation quality assurance, one of the oldest and most popular problem in this category is about the review workflow. Several pieces of literature introduced multi-stage review workflow, which sends an annotation to multiple review steps before approval to minimize inconsistency and human errors. As a result, label consistency and certainty are improved, making the labeled data more reliable. The concept of hierarchical review is extremely relevant for the enterprise AI application where accuracy of label is extremely important to maintain model accuracy and effectiveness.

The cloud based architecture was another topic in the literature. The annotation platform application may be deployed on the cloud and as a team in different parts of world can carry out annotation task. Cloud computing supplies the ways to maintain annotation data centrally, vast bandwidth, supercomputer ability, authentication, online monitoring and boundless extensibility. The new generation of annotation system use the RESTful webservice API, relational database, RBAC and JWT as authentication to expand the business.

To address the above mentioned gaps, combining the following requirements, a complete cloud-based data annotation management platform, namely “Annotix”, is designed in one single application which covers the following functional blocks: dataset management, automatic task generation, annotation flows, multi level quality checking(L1, L2, L3), payments management, role-basedaccess, analysis dashboard, and whole project view. The entire annotation life cycle compressed into a single application is expected to bring significant improvements in annotation performance, quality, human resources, transparency and also enable the extensibility in AI & Machine learning data annotation projects.

## Literature Survey Summary

Above, the state-of-art study shows that annotation tools have become specialized for teamwork, cloud, automatic task distribution and verification. Recent work by Sapien AI (2022, 2023) researches fast cloud annotation pipelines and task distribution systems, and the works by Digital Bricks (2023) and Annotate AI (2023) concern human-in-the-loop verification, workers’ efficiency and elaborate annotation processes. State-of-the-art commercial annotation tools such as Label Studio, CVAT, Labelbox and SuperAnnotate clearly confirm the industry trend.

It is worth noting that the literature survey has also identified that most prior work has only concentrated on providing annotation interfaces, but have not addressed an end-to-end management system, with only limited support for aspects such as project management, multiple level of review settings, global payment and logging system and analysis reports. The research issues and shortcomings identified are then directly mirrored by the design of Annotix as it covers the entire life-cycle of annotation (submit datasets, automatic task generation, monitoring, quality assurance and payments) from a secure and scalable cloud environment.

## CHAPTER 3

### SYSTEM ANALYSIS

#### 3.1 Existing System

Current annotation method on data annotation is exemplified on main-stream annotation platform which provided the minimal set of annotation tools. The popular annotation platform actually excluded crucial annotation tasks that are used for exploring data, such as image, video, audio and text data tagging for AI, ML and CV purpose. Majority of these annotation platforms can only allow the organization to upload/load dataset, establish annotation project or labelling or categorizing tag, and then reduce annotation workload to human annotators. These platforms still remained the main stream on annotation interface.

Many of the annotation tools on the market only have the limited functionality [17]. In the main they are for annotation purposes of images only and contain drawing tools such as a bounding box, a polygon, a segmentation mask, a key point, text classification, an audio transcription and an object tag. They allow you to annotation an image but do not support the management of a large team of annotators [17], breakdown the image into manageable chunks or complete the project from beginning to end. [17]

This is another major weakness, annotation system does not have task management functions. For some systems, a task needs to be broken down into more small annotation chunks by the project administrator, and then send to annotator one by one. For systems with simple task allocate functions, project administrator needs to keep changing the allocation pattern. When the amount of datasets grows up, it will be more and more difficult to allocate equal amount of a task to each annotator. It will cause uneven work distribution, idle workers, pointless task-duplication, long time of project delay... Ineffective task allocation not only increase system administrators work load, but also effects the system efficiency.

The lack of quality assurance mechanisms is another reason for many of the current annotation systems. In one reviews process, that excepts annotation as a final step, most annotation systems do only one review process where the finishing annotation is only reviewed by one reviewer before accepting. Although this review procedure can only afford for certain the annotation errors into processing, it is not valid enough to made fully correct annotation for enterprise level ML train datasets. These annotation errors would finally put into the final ML train dataset within their reviewers' sign-off, which can influence the accuracy and stability of a trained ML. Moreover, these multiple review repetitions like those supported by some present annotation tools are not very well- supported to afford the stable feedback to effectively make correction or fully traceback complete annotation history.

Similarly, using as an annotation tool, it exists only a very few annotation tools able to display the status of several projects or follow a project. For instance, a project administrator should be able to view the task annotation progress, knowing all the performing of the supervisors and the number of hours spent for annotation in the tool and not only the number of annotations and time. If the number of annotations and time are considered, most of annotation tool are not providing advanced dashboard able to show in time the project, annotation quality, workload, performance, completion and cost.

However, the payment and worker tracking is usually done outside of the annotation environment. Most of the annotation systems have no module for payment tracking for annotation and review. Annotation and review organizations have to directly pay the annotation and review workers through external spreadsheet, accounting system, or manual calculation. This paper trail adds the complexity to the organization, provides room for error, reduces transparency within the organization and to individual worker and leaves the room for dispute on payment. Further, the evaluation of the worker productivity, which is critical in rewarding productivity, is also separated from payment.

A major factor is the lack of audit/traceability capabilities. A lot of annotation tools do not have the ability to keep an audit trail of the task a user is carrying out. This can make it very difficult to work out who created an annotation, who peer reviewed it, what changes may have been made and even why a task has been rejected back to comments. The lack of audit trail effects responsibilities, problem management, QA and project auditing for enterprise users dramatically.

Secondly and most importantly, scalability is also an issue for existing annotation as we're faced with modern day AI projects that involve hundred-of-thousand, million, or more data samples, and hundreds-of-annotators-and-players, very few annotation platforms exist that working under that scale. In the absence of automatic partition of annotation samples into independent sub-tasks, when the project administrator cannot wisely schedule the annotation task, or monopolize the management. organizational inefficiency, reduction of productivity, and database efficiency all become a bottleneck in platform scalability.

Security and access control should be improved in most of the approaches. But in some annotation system only minimum authentication facility is offered without the mechanism of providing RBAC fully, secured REST API, secure encrypted communication and centralized control of privileges. This might lead to the problem that some or all the users will be having accesses to projects or data set which might not be necessary for doing their current work. This becomes a security hole for organization sensitive data. A weak support of the granular authorization and centralized auditing logs raised issues of system accountability and secure data handling.

Furthermore, disconnected between the participants in the annotation process occurs frequently. Annotations, reviews, project managers and administrators often have to switch to tools outside CII to-do things such as email, IM, and spread sheets to communicate annotation instructions, resolve review conflicts, or trace collaboration workflow, which not only costs time delay and decreases external productivity, but also wasting replication effort and uncontrolled storage of valuable artifacts in project conversation history.

To sum up, the already exist annotation platforms, as least the minimum annotation map; Making Tool to label the Data Sets for AI and Machine Learning, then, really do nothing but annotation, ignoring all the other critical functions and features along the entire Annotating Project Life Cycle, why such complete full blown Enterprise annotation management system can take shape? Because along these aforementioned missing capabilities, Automated Tasks Management, Multi-layers Quality Assurance, Project Monitoring and Management, Workforce Management, Payment & Transaction System etc. Are still lack of perfect solution. Those aforementioned problems give the answer that an integrated platform is to be a must, which has to be secured, scalable, intelligent, and the nucleoprotein, delay stopping the very annotation process.

## 3.2 Limitations of the Existing System

Annotation tools are available nowadays and they considerably facilitate the process of data labelling than human annotation, however, they are still bound by technical limitations and also practical limitations that affects the speed and also the quality and the scalability of the annotation, which are expected to be higher for thousands of tasks and for dozens of annotators.

Lack of automated task management. All but a few of the systems examined must perform manual allocation of data into tasks and distribution of tasks among the annotators. Within an administration time, this means an unbalanced workload, double annotation and possible conflicts between annotators working on a pair of objects.

The quality checkpoint is usually superficial. Few systems have more than one round of review and it is rarely bidirectional by more than one reviewer for very high accuracy projects, for instance, training set for AI/ML. otherwise, the errors will pass through and contaminate training set. Generally, the methods do not provide any framework nor feed-back mechanism for informed and consistent revision.

The other common gap is that there's no integrated payment solution. A company's payments to their workers are usually done offline through a spreadsheet or an accounting software which is slow and error prone. And, in general there is no detailed reporting for performance either, or at least not through a single interface.

In terms of security and access control it can be missing: It seems not all tools are implemented to be fully role-based authorized on the data and functions and the user has a broader access than they should have with the roles. No audit logging.

Scale has relevance with the scalability of annotation tools as well, since a significant amount of tools was not designed to handle the massive scale concurrency and efficient task scheduling requirements and that presents an unacceptable issue when thousands of annotations a day need to be performed by a massive organisation.

One area where most systems are not very strong is communication and collaboration between annotators, reviewers and administrators. There is usually no facility in the system for annotators, reviewers and administrators to communicate about annotation policies and to resolve discrepancies in reviews. This tends to lead to the teams resorting to e-mail or chat software outside the system which is time-consuming and may lead to inconsistency in annotation decision-making, and difficulty in tracking discussion.

As for the reporting & analytics it's also quite poor. Some system reports simply the count of naive metrics like number of finished and to do tasks. It is very hard to find systems that have an actual dashboard reporting worker productivity and quality, review rate, finish rate and cost. Without such data a project manager can't foresee future bottlenecks in a workflow, nor can a PM can use such data to plan for future projects.

The second issue which limits the customisability of back-end systems, is that a company is almost guaranteed to need to tailor things such as the workflow, annotation guidelines, task allocation and review rules to the domain they work in (medicine, autonomous driving, NLP, manufacturing etc), whereas most annotation systems do not.

There are also many systems currently in place which are not designed in this way for these new AI capabilities which include, AI aid pre-labelling, automated validation, workload prediction and balancing, multi language and integration with cloud storage and which do not have a hardware component to their natively built in software.

Related to this too is data protection: a lot of systems have no backup and no versioning and recovery; if a system were to fail, or a mistaken deletion occurred, then annotation data could be lost forever. This would be an expensive risk to take over large-scale annotation.

Finally most of annotation tools are standalone application and are not part of a larger integrated suite like a project management, billing, reporting and quality assurance tool.

### 3.3 Proposed System

This is the context where a web based platform, Annotix, has been designed, in order to provide solutions for these drawbacks. That is correct, Annotix handles one global, centralized, scalable and secure workspace for the entire annotation cycle, including creation of projects, managing data sets, auto-generation of tasks, annotation, varying levels of quality controls, payment administration, data analysis and monitoring of the project by only one web application.

As the AI/ML training models are getting dependent more and more on labelled data the management of the raw data is always cost-effective when done manually. Annotix end-to-end solution to pipeline annotation from ensuring that data is secure, quality maintained, visibility maintained throughout the pipeline. The system is supporting all sort of data types (Image, Video, Text, and Audio) and all sorts of domain (Computer Vision, NLP, Speech, Autonomous Vehicles, Healthcare, Agriculture, E-commerce, etc.).

Firstly, a company administrator logs into the web application and builds a new annotation project by setting up the annotation type, size of tasks and number of reviews necessary, the payment strategy, instruction for annotators and the deadline for completion. After project is set up, the data is upload via web browser to an authenticated site where it is verified for format, completion and metadata and then input into the system.

Once the dataset has been uploaded. The system would automatically split the dataset into annotation jobs according to the pre-defined task size. Then these divided annotation jobs would be saved into PostgreSQL database with status “Pending”. The system can then automatically split the batch job into individual annotation jobs, eliminating the need to create jobs from the interface directly, and handle thousands of files in a single batch. Sysadmin can monitor all annotation jobs anytime and change the project configuration.

Task Management module: after authentication the annotator can view and claim the desired tasks from a custom task list. As soon as a task is claimed the task lock mechanism will lock the task for the defined default time window (which is defined as the max time the task can be assigned to one annotator). If the task still not finish at that time, this lock will be eased and the task becomes available again to claim by other annotator.

Provides a clean annotation interface for the annotator to provide label for image, video, audio or text following the project annotation instructions. All edits are automatically saved, and then the work would not be lost on crashing, or unexpected disconnect. The final annotations are sent to the backend through secured REST APIs and put in a queue to be reviewed.

Another of Annotix's strengths is its multiple review pass levels. While most annotation platforms have a single pass of review, Annotix provides for a multiple-level of reviews ranging from L1 Annotators up to L2 Reviewers and, if needed, L3 Super Reviewers. Once an L1 Annotator has finished their task, the annotation is then passed onto an L2 Reviewer who checks the work to ensure it is accurate and approves or rejects the work. Approved work can then be forwarded to an L3 Super Reviewer to ensure work accuracy even further. Rejected work goes back to the previous stage of annotation with reviewer comments for the annotator to ensure they correct the work properly.

If it is for watching live projects at first glance then the system has a dashboard for analytics, through which Company managers can access annotation progress, precision, completed jobs, worker productivity, review stats, allocated jobs, payments and deadliness in one interface in order to identify bottlenecks, worker productivity and important achievements.

System security is achieved through the use of JWT based authentication/authorization and Role Based Access Control. The user will be authenticated through the use of login name and password on the platform, then granted access based on the role of the user. Communication between the React.js front-end and Spring Boot back-end will be over secured (HTTPS) authenticated REST calls and input validation, exception handling, audit logging and transaction control will be maintained throughout.

Calculates the workers payment automatically depending on number of task completed & validated - keeps the complete record of workers payments and commissions and can be integrated to any payment gateway for payments.

The Annotix backend is built on the Spring Boot framework, and the UI layer is presented using the React.js JavaScript library. All application data (Users, projects, datasets, annotation tasks, review entries, payment details, logs and audit trail) is stored in one central PostgreSQL database. The business logic is divided into services (modules): Authentication, Project management, Dataset management, Task management, Annotation, Review, Payment, Notification and Reporting.

Its service oriented and modular architecture ensure that the platform will be very flexible and scalable. Future enhancements to be made will be in AI-based annotation recommendation, machine-learning based data quality validation, cloud synchronization, multilingual annotation, proactive labour estimation and advanced reporting all can be chosen incrementally without modifying its architecture to a great extent.

Generally speaking, Annotix offers good authentication, automatic task generation, multi-layer QC, central monitoring and embedded payment system on highly scalable cloud platform, all of which drastically speed up the annotation process and cut down human workload, and elevate data quality as to meet up the rapidly increasing needs of AI and ML development.

### 3.4 Advantages of the Proposed System

All the steps that the annotation lifecycle (corpus ingests, annotation & analysis, figure based design, quality assurance, reporting and invoicing) has been integrated in one cloud based application as compared to classical annotation management systems. It fulfils the exact needs of the annotators and large scale performers who work on conventional systems (e.g. several standalone annotation, quality and reporting and billing applications) in a single application which has already proven beneficial for improving efficiency.

Arguably the most exciting of Annotix's features is that of Fully-automated task management. With virtually every other toolset of annotators currently available, the administrator has to pre-load in data to have split into some number of annotations to be distributed out to annotators; this is usually both tiring and time consuming, as well as the fact that the annotator work load distribution is unequal. Annotix fully-automatically slices through the uploaded data to create the number of annotation tasks the administrator wants with desired parameters (e. G., how many annotations, type of annotation, review workflow, etc) during run time, then automatically evenly divides work between all the available annotators; this reduces administrative overhead considerably and optimizes the time it is truly human-based.

The task locking mechanism in it is another nice feature and a highly effective one. Once an annotator "fixes" a task, it gets locked and disappears from other people's view until the annotator is done with the task, or until the time for a task is expired (annotator is prompted to do just one action, if time is expired, which is to release the task). That means no overlapping work and no conflicts between annotators for annotation and also each data-set will be annotated only once unless the annotation part needs another cycle due to the review phase. If a reader overdoes a task annotation then it returns to the pool.

Perhaps one of the more distinct features of Annotix is its multi-level QA workflow -how it manages to check accuracy and consistency of annotation data. It is understandable that this is a generally flexible feature of annotation tools but the need for several rounds of review is unusual and essentially it is the organization level system that Annotix implements. This system comprises of \*\*L1 Annotators, L2 Reviewers and L3 Sr. Reviewers\*\*. When the user is satisfied with their annotation, they submit it for review to the reviewer who can either accept or reject and add comments (to give instruction for correction, if necessary). If necessary it can be submitted to Sr. Reviewer for review before entry into the final dataset. This multi-level system finds annotation errors and also increases the size and consistency of the final dataset as well as a powerful training data for Machine Learning.

Besides, the features of Annotix' project monitoring and evaluation panels and statistics is quite powerful. Administrator can trace annotation projects in real-time, live. Via interactive panels project monitors can observe the projects at their close examination including the quality control on annotation tasks, investigation on reviewers' capabilities and performance, the tracking of workers' work load and effectiveness, monitoring on payment system, and exploring on the deadline of the projects. Having real-time direct feedbacks of annotation performance, project monitors could timely and swiftly improve the workflow, planning and schedules.

Another positive point of Annotix is its integrated payment system, which will do all the payments to annotators. While the free annotation system does require you to extract the payment data and process it on your own accounting software or in another spreadsheet, Annotix calculates your payment for the annotation workers based on accepted annotation jobs against each batch of payment rules and provides you the full records of payment history and book keeping for each worker and reviewer. Such efficiency can offer incentive for annotation workers to contribute, and avoid calculation errors.

**Security:** Security is another one the most crucial features which is given by proposed platform. It supports authentication mechanism by using JWT authentication, Role based access control mechanism, HTTPS protocol for the REST API's, encrypting all the communications between systems, all requests received are validated at the server end etc, so that Management data can be accessed in secured manner. User's gets access to the system only post they login and a user is authorized by its role (i. E. System Administrator, Company Admin, Senior-Reviewer, Reviewer or Annotator). Besides that, Role based fine-grain control of the functions that can be performed by the authenticated users makes it very hard to misuse of data, since users would not able to access, manipulate plain data sets. It supports logging important system operations.

The third advantage is the software modular design . Annotix has many functional modules which are independent with each other such as Authentication module, User Management module, Project Management module, Dataset Management module, Task Management module, Annotation module, Review module, Analytics module, Notification module, Payment Management module and Reporting services. Each module has a specific role to play, but they are communicating with each other via REST API. The modular design of the system will reduce the maintenance costs, facilitate the debugging and troubleshooting of the application, allow parallel developing process to develop the software, allow module substitutability and up-gradability to achieve easier upgrading.

handle multisource data sets, as it can support images and video, audio and text documents annotation tasks. An organization that performs an annotation task on image and video, audio or text documents in the domain of health care, autonomous driving, agriculture, manufacturing, e-commerce, finance, education, natural language processing and speech recognition industries can employ the same annotation tool.

Additional major advantages lie in data management. Data management all pieces that form an annotation project users, organizations, data sources: data sets, annotation jobs, annotated data, review data, payment, auditor data, parameters is kept in a centralized secured PostgreSQL database. centralized database allows data integrity; stronger back-up and restore capabilities; more efficient (ETL) extract, transform, load processes aimed at enhancing report efficiency related to each of an annotation project element; administrators has a fully access over annotation project elements and tasks; traces are preserved to trace annotate, review, payments and administrator activities back.

This platform is designed to be so **\*\*highly scalable\*\*** that one can launch an enterprise-level annotation project on this. Using recent cloud computing techniques, Annotix can host thousands of users and thousands annotation projects and tens organizations without performance drop and even with increasing size of project and amount of data, the resource can be expanded in real time.

The last area where the Appendices have the most significant impact is on “Collaboration and the resultant workforce productivity”. Integrating annotation, review, project management, payment processing, analysis and communication under one Appended Platform system removes multiple disparate systems. Notifications are automatically generated and delivered via e-mail on new assignment, review comment received, status updates or issues with payments, as soon as they are submitted. Dashboards allow project managers and administrators to stay constantly informed on the overall progress of all projects and communication to increase overall waiting times and ensure that all parts of multi-national annotation team are well-coordinated.

An advantage of the platform is that it can grow with technology into the future. The architecture has been built so it can move forward with the current technology into the future. There could be automatic prompting to the user (based on the machine learning system), automatic checking of the quality of the annotation, predict system workload balancing, make the annotation system multi-language, sync to cloud storage, predictive annotation system, real time analytical dashboards, microservices based deployment.

The system has been more reliable and easy to maintain using the following generally accepted software engineering standards: RESTful interface is being utilized to obtain friendly interface as cooperative communication, where all client’s modules are involved into interaction. Centralized data storage enables system to be consistent in data. But it is possible to easily make it much stable, reliable and maintain an error-free state of system, such as employing exceptions, long transactions, validating and security. Layered architecture enables the system to be maintainable, allowing the programmer to easily debug different components individually and acquire the immediate results.

As far as enterprises are concerned, Annotix saves a huge amount by managing many of the admin functions performed manually, such as task assignment, project monitoring, payment processing, report generation, etc. This way, admins can get rid of their team collaboration, quality control and future project preparation. They can shift to more high-level management and quality assurance of projects instead. Through systematic task assignment and management, it is not hard for enterprises to reach higher throughput and much greater return of investment in mass scale AI data preparation projects.

Summary, in this scheme, a total annotation management platform has been built which could effectively deal with the main defects of the conventional annotation systems including non-scalability, non-remoting, low security level, insufficient HR management and poor quality control. It takes many effective measures including automatic annotation project distribution and evaluation, intelligent annotation workflow and user annotation suggestions, multi-level quality control, integrated browsing and managing annotation project, integrated passing of payment, strong but dependable platform security mechanism, flexible and extensible modular software framework, distributed cloud platform for mass annotation experiment, and abundant collaboration functions in single annotation platform. It also provides a strong technological platform with good stability and clear transparent and scalable platform design to support the next generation of AI, ML, business annotation platform and associated fields.

## CHAPTER 4

### SYSTEM DESIGN

System design is the most critical stage of the software development life cycle where all the functional and non-functional requirements determined in the system analysis stage are transformed into a specific technical design for building the system. System design dictates the architecture of system, the set of software modules and its related software, their interfaces, database schema, communication techniques, data flow etc. in order to establish the system which is reliable, scalable, safe, secure and maintainable. In addition to the above system design focus on building a flexible architecture that can satisfy current business demands and yet flexible enough to extend to future requirements

The Annotix platform is a web-based application on a cloud environment which covers all stages of AI data annotation projects; it is a modular platform designed based on a conventional three tier client server architecture composed by the Presentation layer, Business Logic layer and Data layer. This three layer architecture ensures an improved maintenance, extensibility and quality of software package because of the independence of the layers, allowing to modifying the user application interface without affect the other layers, particularly the data layer.

The presentation layer code base is written in React.js. The user interface will be a web based responsive interface so that all 5 classes of users ( Company Administrator, Annotator(L1), Reviewers(L2), Senior Reviewer(L3) and System Administrator) can access them. Web interface will be used to display projects, create annotation, show review dashboards, show reports, add/update and delete users. Frontend would be communicating with the backend system via REST API calls on http.

**Business Logic Layer** The part makes use of the frame work **\*\*Spring Boot\*\***. The control part of the system is this layer. All business logic services is provided through this layer to platform. All kinds of business rules are defined in this layer: like User Login, project manage, dataset importing, automatically task creation, task assigning, annotation manager, review and payment, email notification, generate report, etc. All the API call are decorated by security rules with authentication by JWT tokens. RBAC mechanism is introduced to make sure the user could only get the access to feature and information only that he/she has required authorization. The business service layer is the control centre of the system making sure all data have frequently consistency, communicate between all modules and all business logic rules be strictly executed.

As for the data layer, it takes a reliable relational database management system called PostgreSQL as the most important component of the data storing structure. The data layer will store accounts information of users, companies, projects data, Uploaded datasets, annotation jobs, annotation result files, review data, billing information, system config data, notification messages and logs data. And data layer also makes the use of relational constraints, indexes and transaction control to ensure the validity, consistency of data, fast annotation record fetching, and multi-user access.

Annotix has been architecture modularly in the sense that every high-level functional area has been encapsulated in an implementation of it instantiating it as a standalone functional software module for the given business function. The modules of the business process are User Authentication, Management of Company, Project Management, Dataset Management, Automatic

Task Generation, Managing of Tasks, Managing of annotations, Multi-Level Review, Payment Management, Dashboard & Reporting, Notification Services, Audit Logging. The modular architecture wherein each sub-system does each business operation but loosely coupled by sharing Rest APIs will lead to high cohesion and low coupling of all modules, easier to debug, test, deploy and add new functionality to the business processes simultaneously from different teams.

Security is a requirement from the design stage and implemented as part of the architecture. Our platform has secure login with JWT token, Secure communication using HTTPS, Role Based Access Control, server side validation of all user input, common exception handling, audit logging and so on to secure sensitive organizational data and project data. All requests made from any client application have been validated and authorized before it can be used to access the protected resource as well as sensitive organizational dataset and project data and providing accountability to the user activity within the annotation life cycle.

The architecture of the system is designed with some or all of the goals: high scalability, high performance, high availability, high redundancy, high reliability and low cost. For instance, an annotated AI project might need to deal with hundreds of thousands of other users and down millions of annotation tasks all at the same time. In the case of load, the system is designed to handle load in various contexts, access load to the database, the usage load to individual services or to the central cloud service. Also, organizations, projects, users and datasets can be added while the original projects remain untouched. Also the storage and computing capacity can be scaled up to handle annotation load in the project.

Overall, the architecture of Annotix offers a strong technical solution in building a secure, scalable, maintainable and performant annotation management framework. Another additional factor, besides the ones already outlined, is the use of up to date web technologies, which is essential in providing necessary performance gains required for largest organizations. Together, layer based architecture, modular software design, secure communications, centralized data and cloud based accessibility enable Annotix system to fulfil the operational data annotation demands for the largest of AI & ML projects, at the same time allowing future development opportunities.

## 4.1 High-Level Design (Architectural)

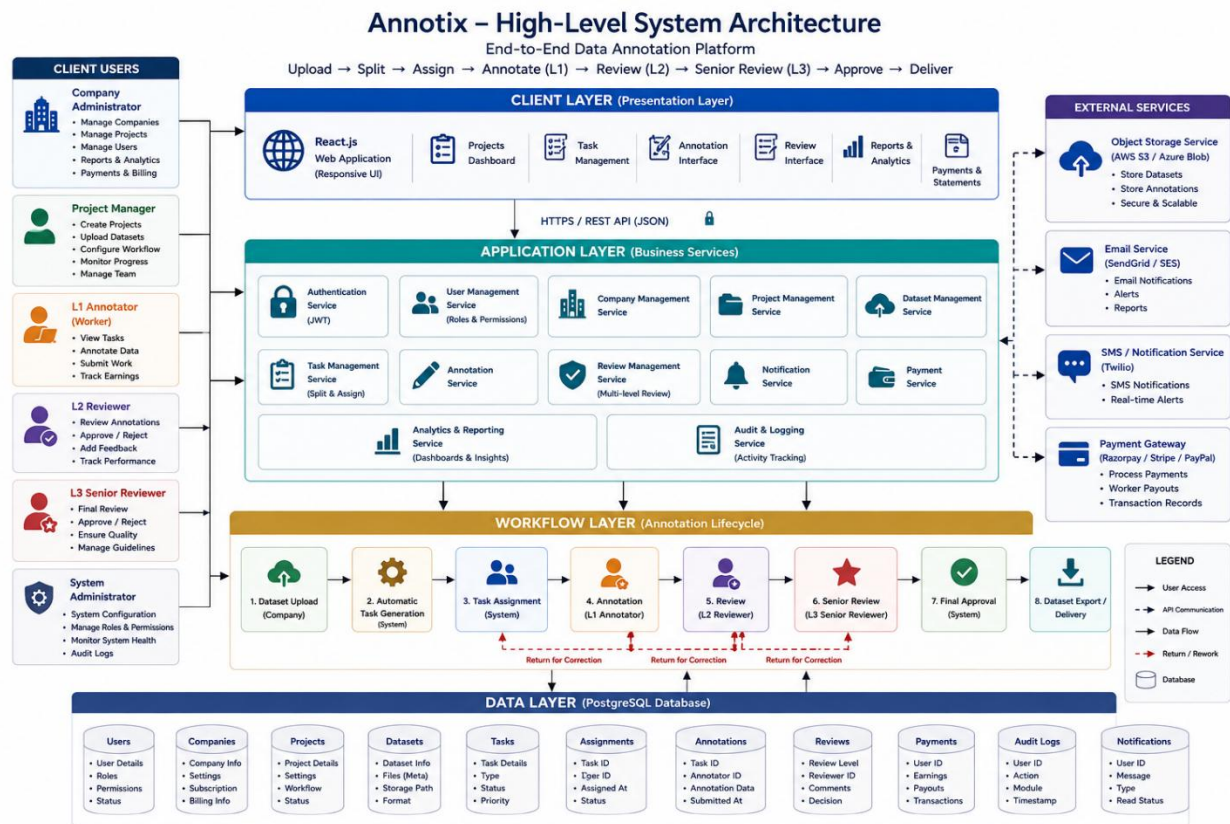
Annotix High Level Design HLD shows the High Level Design of Annotix. It shows the architecture of Annotix and demonstrates how the software components are interacting with each other to produce the complete data annotation application. It shows the various top level modules, various types of user, interfaces between the modules and the data flow between the layers.

Based upon the analysis above, the architecture is determined as below: the layered client server architecture. There are 4 separate layers make up the application: . The role of each layer is to provide the different layers with abstraction service via a reduced set of APIs for each layer. It provides functionalities defined as required for architecture construction, control, test, and extension of platform.

This application was intended to work as an interface for users with roles like \*Company Administrators, Project Managers, L1 Annotators, L2 Reviewers, L3 Senior Reviewers and System Administrators\* differently by granting some privileges to the users of these particular roles. The authorization of the users belonging to the aforementioned roles is controlled using the RBAC

permissions. A web interface has been used to interact with the application and this has been developed using the React.js technology. All the business logics such as authentication & authorization, project handling, dataset handling, tasks assignment & distribution, annotation and review processes, analytics, notification services and payment handling etc. Have been developed using the services created using Spring Boot framework and all the application data has been stored in the PostgreSQL database.

Multi-layer design makes the multi-organization platform, multi annotation projects and hundreds of users be able to run on a single platform without compromising the scalability and security. Secured communication using JWT-based authentication, serving static files, data repository, flat business services, multi-layer object oriented workflow execution, etc. are the features that make the platform scalable, enterprise data annotation project compatible (with AI, ML, etc.) and easy to maintain.



**Fig. 4.1: High-Level Architecture of Annotix**

The Client Access Layer would be the core platform for the end-user to interact with the Annotix system. Currently, the system is just a Web responsive application developed using React. With this kind of implementation, the platform would be used within any modern Web browser in a way where there is no need to install any other application. This implementation would ensure that any user would be able to perform the annotation activity easily from any device such as a desktop, a lap top or even a tablet.

Furthermore, the architecture is prepared for a future extension of the platform. In this moment, we only have a web platform, but this framework offers the possibility of a native app extension for Android and Ios. In addition, Annotix supplies the organization with a secured RESTful API, through which they will be able to integrate the Annotix platform into their current

Artificial Intelligence/Machine Learning pipeline and automates the dataset uploading, annotation and projects procedures.

The system's functionality is controlled by Role-Based Access Control (RBAC). Every logged-in user has access to a set of modules and actions depending on his role. Company Clients are Users who can setup annotation projects, upload datasets, define annotation parameters, monitor annotation project progress, download annotated datasets after validation. L1 Annotators are Users with annotation permissions who can claim annotation tasks that are available, label data samples and submit annotated data for review. L2 Reviewers can examine annotation results submitted by L1 Annotator, check whether they are valid or not, and either approve or decline submitting annotation results with comments. In case the annotation project requires further quality check to be performed by an additional group of users (L3 Senior Reviewers), the annotations will undergo a second review performed by those users, so they can finally verify the annotations given that their previous results have already been approved. System Administrators are Users with permission to manage users, organizations, annotation projects, payments, system settings and monitor system behaviour.

### Application Layer

The Application Layer specifies the business logic of Annotix and is developed using Spring Boot, offering a set of loosely coupled services which communicate through well known REST APIs. Each service exposes specific business functionalities.

**Authentication Service-** For registration, login, check password, JWT Authentication and session management. Authentication service will be called before every request from client application to check for the access to resources.

The User Service is responsible for customer information referring to user profiles, user organization, user position and authorizations. The customer information is also related to the customer account management including role management and access control over the platform.

The Project Service manages the entire annotation projects from the creation of the project, to its configuration (price model, annotation parameters, review level needed...) and the project management.

The Task Service that is responsible for dividing the uploaded datasets of a project into small manageable tasks automatically if it is demanded by the project configuration. This service is responsible for task assignment, task claiming, task locking and task status tracking.

Additional backend services enhance functionality of the platform. Workflow service enforces business rules over annotation lifecycle regarding annotation and review states. Annotation service stores annotation results, maintains annotation history, supports autosave during annotation activity. Review service stores reviewer decisions, comments, approval or rejection of annotation, quality metrics over annotation for each review process. Payment service handles calculation of earning for annotators and reviewers as well as commission and payout amount automatically. Notification service generate notifications via system, email, in-app messages, or SMS regarding task assignment and review result, payment details and project updates. Reporting service provides

dashboard, project status report, annotation reports, quality metrics and analysis for administrators and project managers.

As each business capability is effectively implemented as a separate service component, we can in the future maintain, diagnose bugs, test, deploy and extend the system without disturbing other unrelated modules.

## External Services

Annotix interfaces with outside services in order to provide additional capabilities as well as facilitate enterprise deployment scenarios. Uploaded sets can be stored within EEP, either internally or externally through the use of services like Amazon S3 or Google Cloud Storage where signed URL's allow access to data without the user gaining direct access to the storage infrastructure.

Provides a plug-in to interface with Email and SMS gateway services for the generation of notifications about new task assignments, review feedback, project updates, payments made and received, and other essential events which take place on the system. These may be used by administrators, annotators, and reviewers to coordinate their work more effectively and in a timely fashion.

Financial transactions, Annotix allows linkage with main gateways payment (Razorpay, stripe and PayPal) in order to allow payments of worker wages, commission, and organizations payments. It will allow easy and secured all payment transaction handling.

## Workflow Layer

Workflow Layer Manages the entire annotation lifecycle and controls the flow of datasets through various processing stages. Once a dataset is uploaded by a company, the back-end splits it into various annotation tasks according to the configuration parameters of the project (type of annotation, size of task, need for review, etc.), stores them at a central location, and makes them available for assignment to competent annotators.

L1 Annotators select the tasks from the task pool, do annotation tasks and then collect all the completed annotation work and deliver for quality check. The annotations delivered from L1 to L2 Reviewers will be checked for the correctness of annotation work, and then either accepted or rejected. The annotations accepted will be delivered to L3 Senior Reviewers if enabled by project.

The system manages several workflow rules by default to facilitate the annotation workflow. For example, the task-locking prevents duplicating effort from multiple users, where tasks are locked in the system by an annotator once they have taken ownership of the task. When an inactive timeout value is exceed, the task is automatically released back to the task pool. Some workflow escalation rules are also implemented to immediately escalate a request to a higher level review if the reviewers takes more than the specified time limit to submit a review. Workflow rules like these, which are stored centrally, allow administrators or super users to optimize annotation efficiency while producing consistent annotation quality by reducing bottlenecks caused by the human in the annotation cycle.

## Data Layer

Data Layer - Data is persisted in a large-scale central relational PostgreSQL database. It is composed of Users, Projects, Datasets, Annotation tasks, Task assignments, Annotation results, review logs, payment transactions, audit logs, notifications, and system configuration information, which are all persisted into the Data Layer. Referential integrity and indices are provided, along with optimized querying for reporting.

All necessary information like uploading data, assign jobs & tasks, annotate, judge the judgment, payment, operated by user, system administrator etc. Could be saved in database since its traceability and audibility.

## Security and Future Scalability

The entire architecture is secured at all the layers. All client requests first gets validated by Authentication through JWT based authentication, then all the clients requests gets to access the back end services. Access Control using RBAC is applied so that user can access the module/resource based on their role, then only allowed resources is accessed by clients using HTTPS and for server side we can do the input validation. All the logs are maintained at a single place.

This flexibility of the architecture also lends itself to extending functionality for the future, e. G., it can be easily extended with system suggested annotations, analysis quality verification, multiple language support, live analytical reports, running on a container using docker or a container orchestrator like Kubernetes, or moving to a microservices architecture, without needing to refactor any of the already developed services.

In summary, the HL Design produced an efficient, safe, expandable, and manageable architecture for Annotix. Coupled with an adaptive web UI, verified login, modular services in the backend, a single source of truth for workflow management, incorporation of external services and a robust PostgreSQL database; a system capable of covering every aspect of an end-to-end modern AI/ML data annotation project was constructed. In figure 4.1 we see how the Client Access Layer, Application Layer, Workflow Layer, External Services and Data Layer all converge into a single, cloud-based annotation management system.

## 4.2 Low-Level Design

The High-Level Design (HLD) gives an overview of the architecture and components of the Annotix platform. While High-Level Design is concerned with platform architecture, the Low-Level Design (LLD) is concerned with each module's internal implementation and interactions, database access, workflow execution, API interaction. It is a detailed representation of how the platform would process user requests, including how data would be stored and retrieved, how business rules would be implemented, how the user-request would be coordinated with the overall system by interface components, backend components and the database. The Low-Level Design gives the outline of the implementation of the framework providing details of the different modules, the interaction of services, the handling of the request, the transaction in the database, and the workflow execution.

There is a modular service-oriented architecture in Annotix. That's means each module is handling some specific business domains, and each module can interact with other modules through

STOMP messaging or secured REST APIS provided by the spring-boot application. Such architecture allows loose-coupling and high-cohesion as each group of a module encapsulates its own specific business logic, which provides a smaller surface for each module and it simplifies the development, testing, deployment and maintenance.

### Authentication Module

The execution flow of our system starts with Authentication Module which is the entry point for each user having the system. When the login credentials are submitted through React.js frontend by the user, the Authentication Service checks the Username or email address and matches the provided password with the stored encrypted password through secure hashing algorithms in the PostgreSQL database. An authenticated JSON Web Token with the username and role of authentication user is then generated.

This created token is then send back to our client application and embedded in each authorization header for each API call. Before any call is handled by our backend service our token is validated before we check its validity among other checks. With the help of RBAC the user can only perform actions and request system resources that are granted to his/her role. The roles mentioned during our role validation are Company Administrator, L1 Annotator, L2 Reviewer, L3 Senior Reviewer, Project Manager and System Administrator. Unauthorized call is declined at this point.

### Project Management Module

Once the Company Admin logs into the system, they will login to the Project Management Module which manages the entire workflow of annotation projects. In this module the Company Admin can configure a new annotation project, update parameters for annotation, configure payment mechanism, define categories of annotations, configure size of tasks, configure review level, configure schedule for projects, and assign annotators/reviewers to a project.

The Project Management module validates any project setup before saving the project information into the PostgreSQL. The project management module keeps a list of different types of metadata for a project, e.g., project ids, project description, annotation type, workflow configuration, payment configuration, review configuration, task configuration, due dates, project state. Based on this setup a dataset processing task can be defined.

### Dataset Management Module

After a project has been created, the project administrators may upload datasets via the Dataset Management Module. This module verifies the file types allowable to upload, file size constraints, metadata consistency and project scope before uploading dataset files. Dataset metadata is stored inside PostgreSQL, while dataset files are stored either inside the local storage or in external Cloud storage services depending on deployment configuration.

Dataset Management Module's fields include dataset ID, dataset project name, dataset upload date/time, dataset storage location, dataset file format, dataset file size, dataset processing state and ownership attribute. After passing the validation, the dataset can then been automatically used to generate jobs.

## Task Generation and Task Management Module

After data validation has been successfully completed the data is passed to the Task Generation Module in order to: The task generator receives the dataset and generates automatically some annotation tasks by dividing it into batches based on some configurable criteria. E. G. The size of annotation task can be specified on the basis of the size that is requested for generating the project. This is another way to online divide the annotation task instead of having it manually designed.

Afterwards all the tasks are to be inserted to the PostgreSQL by having ‘pending’, the taskid, projectid, priority, created on and the assignee.

The Task Management Module takes care of the complete Task life-cycle. This Module is in charge of managing the Task queue (new Task’s) and the free annotators (new Task’s to perform), besides monitoring and keeping track of the state of a Task at any particular time during the execution and the state of the tasks at this time. When an annotator is about to take on a task, a task locking mechanism is triggered which essentially blocks this Task for this specific annotator.

Flexible time-out settings are configured for this module to ensure smooth progress of the workflow. If the task was assigned to annotator and the annotator fails to submit it back to system within the assigned time frame, the task will be automatically unlocked by the system and pushed back to the pool of ready tasks for other suitable annotators to pick.

## Annotation Module

Annotation Module is the fundamental component of the Annotix platform which allows annotators to execute labelling tasks. After a task is allocated, users utilize the annotation interface from React.js front-end and carry out the labelling’s based on project annotation guidelines.

To ensure accuracy, as well as user convenience, the Annotation Module allows for auto-saving of annotation progress. As annotating is performed, progress is saved, and any annotator can later pick up a task at the last point of progress saved.

Once the annotation is submitted it is checked and committed to the Annotations table in the PostgreSQL database with annotation data, annotator information, date/time, version history and submit status. The task status is also updated from In Progress to Submitted for the quality check.

## Review Management Module

The Review Management Module provides the core hierarchical QA paradigm for the platform. An incoming annotation is added to the review queue, where the annotation quality has been checked by a L2 Reviewer who reviews the annotation for correctness as specified by project guidelines. L2 Reviewers review submitted annotations, look for appropriate labelling, and either approve or reject the job.

In the case if the submission is rejected by a reviewer, the review comments and the rejection reasons are recorded in the database. The task is then sent back to the original annotator with the preserved review history.

For projects that demand more rigorous quality check, the L2 approved annotation are sent to the L3 Senior Reviewers as another review stage. The Senior Reviewers do the final quality check and approve the annotation before it comes to the final completed data.

All review actions, approvals, rejections, reviewer comments, quality scores, review times, previous review actions, are recorded indefinitely, providing a full audit trail for each annotation.

### Payment Management Module

Payment Management Module handles all the payments related to the annotation tasks. Payment to workers is automatically determined by the approved number of annotation jobs, project specific pricing, pay scheme, annotation complexity, review results and another payment scheme rules.

All the information regarding the payment is available in the system at a glance such as the details of the earnings of an annotator, the salary paid to the reviewer, commission paid, history of payments, details of each transactions, payout schedule, report of payments for organization etc. The payment related information is stored in the PostgreSQL database and payments are made via gateways such as Razorpay, Stripe, or PayPal.

Auto-calculated payments can greatly diminish administrative burden by streamlining processes and add to the transparency and precision for organizations and annotation workers.

### Project Monitoring and Reporting Module

Project Monitoring and Reporting Module is provides administrators access to the current state of all annotation projects by extracting working data from all other modules. It surveys the progress of tasks, annotation coverage, reviewer productivity, worker output, quality, overall project advance, payment statistics, workflow status to visual charts.

With such reports, project managers can track the current status of a project in real time, bottlenecks in the workflow, efficiency of the annotator, evaluate reviewers' performance, and reallocate workload in an informed manner and reschedule the project.

### Database Communication and API Processing

Interaction between the React.js front end and back end is restricted to secured REST api calls, created in spring boot. Each rest call is made over https, authenticated via JWT and authorized based on the role assigned to the user.

Before the execution of any operation, every request received passes through the authentication, authorization, request validation, business rule validation and database transaction before being carried out. Passes the request to backend service for altering the PostgreSQL database via create, read, update or delete operations.

The relational database keeps all the persistent application data, such as user information, user profiles, organizations, projects, datasets, annotation tasks, task assignment, annotation results, review history, payment transactions, audit log, notifications, workflow setup and system settings. Parameters of relational and transaction consistency guarantee the persisting of data and thus facilitate reporting and analytical queries.

## Extensibility and Future Enhancements

Thanks to the modular implementation, the Annotix platform can enhance progressively without major architectural changes; all business function traits are encapsulated into autonomous services, thus all future improvements from AI helpers to annotation suggestions, to auto validation and quality control via Machine Learning, to multiple-language annotations, to advanced reporting, to prescriptive task distribution, to cloud native (docker/k8s based) hosting, and finally to microservices architecture can be added over time without breaking the current modules.

The Low-Level Design in general offers a detailed approach for the implementation of Annotix and describes the interaction of the services in a detailed manner, such as the connection between authentication, project management, dataset processing, task creation, annotation, review, payment & reporting and database services. When considering a modular structure, secure REST API communication, a centralized PostgreSQL database, your automated methodology for project flow control and a service oriented architecture, this Low-Level Design is able to serve as a solid, scalable, maintainable, and secure foundation to run enterprise scale AI and Machine Learning data annotation projects.

## CHAPTER 5

### DATA COLLECTION AND PREPARATION

Data collection and organization is the basis of any annotation system built in Annotix. The quality, organization and validation of source data are a crucial part of the quality of the end result an annotated dataset. While machine learning tools are mainly used for model training, Annotix only deals with dataset management during the annotation process, making possible dataset input, validation, task creation, annotation, review and output. Attention is therefore driven to robust dataset management, appropriate data organization, preprocessing and automation of dataset preparation so that annotation teams may work with organized, validated dataset inputs.

The increasing demand of Artificial Intelligence (AI) and Machine Learning (ML) applications creates a huge need for handling big data. In many companies there are dozens or thousands of files coming from different sources, often stored on different systems and in different formats. Constraints and human resources, on the other hand, make handling the data sets on an excel sheet, on shared folders or in personal email accounts impossible. The files are duplicated, retrieved with wrong or inconsistent file names, the annotation data has inconsistent or empty meta data, the database is corrupted with missing data, and tasks are assigned inefficiently. Annotix mitigates the pain points of data collection by providing a central cloud-based platform to normalize the collection and preparation of data for annotation.

Annotation six was created for various use cases a single platform can support many common input formats found in an AI Project.. Images, video, audio and text data can all be worked in the same annotation flow. Having various input formats supported in the same annotation flow allows customer to use one platform for any of their needs in a broad range of industries such as computer vision, NLP, speech recognition, healthcare, autonomous driving, agriculture, E-Commerce, manufacturing or education without using various annotation systems.

Once each of these data sets has been uploaded, the Dataset Management Module will interpret and run automata tests on each upload before allowing them to be accepted into the AT data management system. The functionalities implemented into the validation routines will permit useful, complete and supported data set to be imported into the AT's annotation system.

The validation which will be performed will include: supported file formats; the size of the file; whether or not the file has finished uploading. It will verify presence of all metatags that are necessary. It will also check for duplicated files in the project and possible duplicated records, which will be flagged.

In addition to the validation of the uploaded dataset(s), Annotix also records a number of vital data associated with each dataset, and then uses this as metadata about the dataset within a central database. Data is stored in the database as metadata including: ProjectID, DatasetID, Filename, Filetype, Upload time, Storage Location, File Format, Dataset Dimension, Category and status. Centralising project and dataset information in this way allows easier management of projects, faster searches and reporting and will provide full traceability.

Once the uploaded dataset has been validated, Annotix automates the creation of annotation tasks through the automatic instigation of the task generation. This instigation of tasks replaces the need for the administrator to manually allocate datasets to annotation workers. The datasets are automatically broken down into a number of annotation tasks, based on the task size specification set when the project was created by the organisation. For example, if the organisation uploaded a 10,000 image dataset to Annotix, and then specified a task size of 250 images, then 40 separate

annotation tasks would be automatically created. All annotation tasks would automatically be created in the default Pending state, and stored centrally for the human annotator to undertake.

Auto task generation reduces the administration workload has the effects of handling the load balancing of the annotation team and also by means of auto creation of annotation tasks. The creation of annotation tasks is done automatically as per the configurations done for project setup. The annotation project of any size can be handled efficiently for the organization.

For each task generated, Annotix manages a large set of metadata for all tasks. Metadata for each task may include project, dataset, instruction, Priority, state, instructed in or not instructed out, workflow state, comment, time entered, time completed, progress and history data. A large set of these tag enables administrators or project managers to monitor the annotation progress, measure the productivity of various work groups or different project and extract operation result report from database.

Data validation is also one of the most important part of preparation phase. Since the platform is doing validation before doing any annotation task for unknown file types, damaged media, unfinished upload, duplicate data sets, invalid project combination, missing meta data, invalid file names etc. This issue can be detected before the annotation worker starts work and save the time of work in poorer data, so that the quality reviewers only left with the annotation quality checking but not the data quality.

Security has been incorporated in during each part of the data collection/preparation, and all dataset upload functions were conducted within an authenticated user session (by applying JSON Web token based security). Dataset management features were protected by RBAC security, in order to support role based management of datasets, to only empower the right to upload using the authorized personnel. Company Administrator and selected team members role could upload, edit, and manage project datasets. Annotators and reviewers who have assigned tag tasks and annotation work could access their respective tags and annotation work. RBAC Security have provided the defines layer for the company CIA of sensitive data, while still able to provide access rights for annotation life cycle. The raw datasets received will be persisted in the platform storage infrastructure, in a at rest condition, and a unique piece of associated metadata will be maintained in the PostgreSQL database. Depending on the multiple levels the organizations choosing to operate with the platform may decide to keep the datasets on a external cloud storage facility or also have the data locally stored but the all will have the same one level of metadata in order to enable all the elements to find the datasets without exposing or covert the system happening underneath.

In addition, Annotix is scalable at the enterprise level. A company could easily have millions or billions of data samples running through many projects at one time. With a centralized data sample organization, an automated annotation metadata management system, annotation task creation interface, and an efficient database structure, Annotix can perform on large enterprise projects with no performance decline. In addition, each organization is isolated within a domain: datasets, annotation tasks, and organizational data is fully segregated from any other organization.

Another aspect that comes up is the platform's ability to do a full audit trail on the data handling. All the operations that are performed like upload a dataset, execute validations, update metadata, create annotation tasks, perform admin tasks etc are all log in to the system's audit trail. The admins can follow the process of the dataset, identify the root cause for issues in the production, check the annotation workflow errors, prove and fight against annotation quality issues and so on. A full audit trail is critical in enterprise setup.

Future enhancements: The architecture is envisioned to scale with larger storage and preprocessing requirements. With the increasing size of a project it may require increasing the

available storage, employing a different cloud storage service, or adding more preprocessing pipelines, or having the automated media processing service be part of the main dataset management architecture.

To conclude, this approach in collecting and preparing dataset adopted by Annotix provides a well-organized, trustworthy, and extendable method of preparing datasets before sending to annotations. Its core elements including centrally managed datasets and annotation projects, automatic dataset validation and normalization, content metadata extraction, automatic intelligent task assignment, rich audit logs, role-based access control, NFS mount on scalable, which play an important foundational role for annotations, reviews, Q&A and delivery phases. Since only verified and well-organized datasets flow into annotation pipeline, the operational traffic would be reduced, annotation volume would have an abrupt increase over prior time, the operational errors would decrease, annotation quality will be elevated, the final goal for annotation project on Artificial Intelligence and Machine learning will come true.

## 5.1 Data Sources

Annotix accepts datasets from a range of sources to support the annotation needs of AI and ML projects, validating and storing each securely before it enters the annotation pipeline. The supported data sources are summarised below.

| Sl. No. | Data Source               | Data Type                  | Description                                                                   | Typical Use                                                        |
|---------|---------------------------|----------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 1       | Company dataset upload    | Images                     | Image datasets uploaded by companies through the web portal.                  | Image classification, object detection, segmentation.              |
| 2       | Company dataset upload    | Videos                     | Video files containing multiple frames for annotation.                        | Object tracking, action recognition, surveillance analysis.        |
| 3       | Company dataset upload    | Audio files                | Speech recordings and audio clips.                                            | Speech recognition, speaker identification, audio classification.  |
| 4       | Company dataset upload    | Text documents             | Documents, sentences and textual data.                                        | Text classification, sentiment analysis, named entity recognition. |
| 5       | Public datasets           | Images, video, audio, text | Publicly available benchmark datasets used for research and testing.          | Model training, evaluation and experimentation.                    |
| 6       | Cloud storage integration | Images, video, audio, text | Datasets stored in cloud platforms such as Amazon S3 or Google Cloud Storage. | Large-scale dataset management and secure access.                  |

| Sl. No. | Data Source             | Data Type           | Description                                                                                          | Typical Use                                                 |
|---------|-------------------------|---------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 7       | Enterprise applications | Mixed data          | Data generated from business applications, healthcare systems, IoT devices and e-commerce platforms. | Domain-specific AI model development.                       |
| 8       | User-generated data     | Images, text, audio | Data collected through mobile or web applications.                                                   | Real-world annotation projects and ongoing data collection. |

The main data source in Annotix is the upload of company datasets; companies upload datasets which they already hold directly to the secure web application. The datasets could be images, videos, recordings or texts assembled for a particular business purpose. During upload process each dataset is linked to a project, checked whether it is in the supported format, whether it is complete, saved in the database and, then dataset annotation tasks are created. This way the organizations retain total ownership and control over their project data.

Besides local datasets tailored for specific organization, Annotix also enables usage of public benchmark datasets. Available for free, the benchmark datasets are used for research, experimentations, algorithm testing, annotation workflow testing before use on production data. The use of public datasets enables research institutions and academia to test annotation quality and annotation workflow efficiency on standard datasets.

As for enterprise-level deployments, the platform provides the ability to merge with cloud storages like Amazon S3 and Google Cloud Storage. Instead of having big datasets stored in the application server, big datasets can be retrieved from cloud storage securely via authenticated storage connection, advancing scaling up and eliminating infrastructure constraints. Large annotation projects can then be managed with millions of annotation samples.

In addition to the zero effort datasets, Annotix is capable of handling enterprise business application data sets. These include data sets from healthcare information systems, manufacturing systems, the internet of things, autonomous vehicles, financial applications, and e-commerce systems that are specific to an domain, as well as the infrastructure for combined usage of business data.

Another commonly used source of data is data from users that can be in the format of images, text, audio or multimedia file that has been obtained over a mobile app, website, held with a customer or digital store. These types of data are thought to be very useful in real-world use cases of AI that are found to be generated repeatedly so they need to be labelled as they go to help the model learn better and dataset stay relevant for use.

No matter which resource a dataset is imported from, a common set of procedures is performed on each dataset prior to ingestion into Annotix. This includes data validation, file-type validation, metadata extraction, duplicate record detection, file upload integrity validation and file upload completion validation. Once the dataset is validated it is fractioned into the annotation / proofreading task set for that project automatically.

Providing an unified unified data repository for enterprise A centralized data repository enables high-quality annotation for industry-wide sectors such as healthcare, autonomous driving, agriculture, manufacturing, financial services, education, E-commerce, computer vision, speech recognition and NLP thanks to support for various annotation types. A unified data repository results in standardization in data collection procedure, increases data quality, provides consistent management of annotation workflow and simplifies project management.

## CHAPTER 6

### EXPLORATORY DATA ANALYSIS

In the scope of Annotix, Exploratory Data Analysis (EDA) entails the systematic exploration and analysis of submitted datasets, annotation pipelines, project status and workforce productivity before and during the annotation activity. Conventional machine learning workflow often use exploratory data analysis to analyze properties and distribution of features as well as dependencies between features in the training data. In Annotix, we perform project-oriented exploratory data analysis to assess dataset completeness, proper alignment and validation in preparation of annotation. The overarching purpose of Annotix is to enable project administrators and data labellers to gain meaningful overview on data quality, annotation progress, workforce productivity and project status in the lifecycle of annotation.

The exploratory analysis takes place directly after users upload their datasets to the system. Project managers and system administrators administer the exploration through the uploaded dataset for quality assurance at each file level and project level. Details such as dataset file size, the number of uploaded files, supported file format, uploaded status, annotation category, project category, and presence of metadata are available for view via the admin interface. This inspection before annotation allows the project manager/administrator to coordinate the data flow to annotation only after satisfactory quality assurance.

EDA is beneficial as well to assist with project planning and resource estimation. Upon completion of data set validation, Annotix generates overview statistics that are useful to an administrator when trying to determine the total annotation effort required by a project. The overview statistics (e.g., the number of data items, the number of annotation tasks, the number of configured annotation task units, the estimated project duration and the estimated project staffing level) enable a project manager to come up with an order-of-magnitude estimate for how many annotators and reviewers will be required to meet all of the demands of data annotation. The workload planning calculations result in evenly distributed annotation tasks over independent annotator schedules thus minimizing downtime and increasing the overall project plan efficiency.

For every analytical dashboard and analytical report presented on the platform, it gives the meaning of project data using figures, descriptive statistics and tabulated report. The dashboards represent many status such as the composition of the data set, the stage of annotation, the rate of task completion, annotator's performance, reviewer's performance, review status, the project timetable, the number of remaining task, the number of approved annotation, the number of rejected annotation and the health of the project. All these dashboards are constantly updated in the real-time when users are using the platform, this helps project managers to visualize the current status and problems in performance in order to give urgent solutions.

Monitoring annotation progress is one of the main goals for exploratory analysis in Annotix. Since Annotix monitors all the annotation activities from creating annotation to submitting, the platform offers an administration a fully picture of annotating tasks currently. Administrator can check full details about each annotation task such as its current status (Pending, In Progress, Submitted, Under Review, Approved or Rejected) which helps the administrator to have an overview of workflow, follow up on the overdue annotation tasks and redistribute them to ensure all annotation works are timely.

The EDA can go a step above even that by analyzing annotation quality analysis via reviewer completion in the multi-level review process. Reviewer approvals, rejections, correction volumes, annotation revision, reviewer comments, annotation correctness information can all provide insight

into annotation quality at a very high level. Administrators reviewing annotation quality metrics can see a high level overview of annotation problems, revisit annotation guidance if need be, and determine if annotation team training/education is truly necessary. Thus, by monitoring annotation quality, the annotation's contribution to useful datasets for Artificial Intelligence and Machine Learning can be improved.

Exploratory data analysis. Besides that mentioned above, workforce performance analysis is also an important part of exploratory analysis. The Annotix system records operational statistics of all annotators, reviewers and project team members, including number of finishes, individual annotation time, review tag completion time, acceptance/rejection ratio, load balance and overall work efficiency. These workforce performance statistics allow the system administrators to utilize comparatively, use incentive strategies to motivate rivals, to improve weak users to upgrade, reallocate tasks balance, etc, that could reflect a much more just decision.

Second, because there are various annotation projects supported by Annotix, organizations can even perform cross-project analysis across various annotation projects. Such comparison of multiple projects includes project rate of production, annotation quality, review acceptance/rejection, labor force productivity, annotation time, cost, etc. With such comparison, organizations can learn from similar project having been completed successfully, optimize resources usefulness, come to more reasonable planning of later projects, and ultimately realize annotation process management standardization in cross departments or cross clients.

Historical reporting: A third component to the explorative functions of the platform is the availability of historic data. Annotix records and stores data on the annotation processes, review results, project deadlines, invoicing and statistical indicators of the workforce performance. Historic data enables the project manager to extract information from previous projects in order to plan the human and time resources, improve the annotation protocol and account for the costs of future annotation projects.

Financial information can also be introduced to exploratory analysis through payment information on the projects. Project admins are capable of seeing the annotation and reviewer costs, as well as if the payment is being made or not, and worker payment rates and project costs. Taking into account financial measurements with annotation measurements helps the organization determine project efficiencies, estimate project cost, and optimize worker quantity in the project without running over budget.

The one single PostgreSQL database which every exploratory report builds upon provides for a singular, accurate, consistent and dependable stream of data which is always up-to-the-minute. As all aspects of every annotation project operation, review, bill, workflow event and administrative process are recorded in that single data stream, each analytical dashboard shows the current state of the system at any point in time. Billing, graphical, and statistical reporting become unified for monitoring accuracy and providing full auditability to the annotation workflow.

Overall, the Annotix Exploratory Data Analysis functions help to provide a top-to-bottom perspective to the administrator of every piece of a data annotation projects from the readiness of a dataset to the ease of a data annotation tool, to the monitoring of its progression, to the tracking of data annotation quality and the productivity level, to financial accounting and a bird's-eye view of the health of the data annotation project. And, the Annotix platform offers such a web-accessible cloud platform, where all these data annotation project management tools and quality assurance instruments, along with the integration and traceability of workflow and audit trail are simply another login into the web, so that you can succeed with large-scale AI and Machine Learning data annotation projects.

## 6.1 Data Visualization Techniques

Visualization of data is a key aspect of Annotix, allowing presentation of large quantities of annotation and project data as graphical representations to facilitate monitoring, analysis and decision making. Given the life of very large enterprise annotation projects where there can be thousands of datasets, millions of annotation jobs, multiple organizations and hundreds of individuals logged in concurrently, visualization of operational information in the form of charts, dashboards and visual reports ensures administrators are able to get an immediate sense of overall

project status, without examining individual database tables or records. Annotix draws away from static numerical output and interactive visualizations portray project status, level of annotation progress, workforce utilization levels, quality and financial performance.

Annotations analysis differs from traditional Exploratory Data Analysis of machine learning datasets, in that what is being visualized in Annotix is not the collection of data analysis of datasets, but rather operational analytics at the project level. To do this all data generated during the course of the annotation project from dataset uploads, to task deployment, to annotation and review progress, all the way through to payment and user activity is continually collected, aggregated, and visualized through dashboards that automatically update as the project progresses.

The Administrator Dashboard is central to Annotix's visualization features. It gathers and displays project data from several of the backend modules in one place. Because all project data is stored in the backend PostgreSQL database, all the elements on the dashboard are constantly being refreshed to represent the current state of the system. When data on the dashboard is analysed and, for instance, the cartogram created, a new data table generated, or the summary statistics constructed, the update is immediately reflected on the dashboard. From the Dashboard, project managers can observe ongoing annotation projects and be alerted to potential problems in real-time.

The Bar Chart is the most widely relied upon visualization technique, which is also often used in Annotix in order to perform comparison over several project statistics. Bar graphs are used in annotations to compare the quantity of annotation tasks among several workflow states which includes Pending, In Progress, Submitted, Under Review, Approved and Rejected for projects. This graphical comparison of project statistics help the administrator to identify bottlenecks in projects, review annotation progress, compare percentage completion of several projects, observe reviewer productivity rate, and measure labors productivity rate. Bar charts provide a simple and easy comparison of several statistical values of workflow statistics and help administrator to find imbalance in workflow and take appropriate actions accordingly.

In the second screen of Annotix, the 2<sup>nd</sup> visualization, Pie Chart is used to compare the percentage of all classes in project data. Pie chart is good in representing the composition of dataset, annotation category, project status, reviewer/User performance, task distribution To managers, they may want to compare of image/video/audio/Text datasets being submitted into Annotix system. And for each of the projects they launched, they may want to compare the percentage of approve/disapprove status for annotation tasks.

Line Charts- Line Charts reveal trends over the duration of your project. Annotix offers charts showing daily, weekly, and monthly fluctuations in annotation, review activity, overall progress of a project, annotation and approval rate, task completion rate, and payment distribution. With this trend you can learn where activity has fluctuated, past performance levels of workforce tasks, lead time on task completion, and predicted resource demand in advance. You can also project your past trends from other projects.

One other feature of this is the Dashboard Summary Cards. These are count indicators that are used to represent the overall health states and summary statistics of operation high-level information. The dashboard summary cards include the number of uploaded datasets, the number of active projects, the number of registered organizations, the number of active users, the number of current annotators, the number of waiting tasks, the number of completed annotations, the number of accepted reviews, the number of rejected submissions, the total amount paid, as well as the overall percentage of project advancement. These reports are surprisingly placed right on top of the dash-boards, enabling administrators to get a glance at the overall health condition of annotation system without having to dereference any in-depth statistical reports.

In order to reduce the quality assurance effort, Annotix offers specific Review Quality Dashboards which intuitively show review behaviors in the dynamically multi-level reviewing process. The dashboards include reviewer approval rate, reviewer rejection rate, correction time, average review time, review workload distributions, annotation quality trend, etc. Under the supports of hierarchical reviewing process (e.g., L2 Reviewers, L3 Senior Reviewers, and etc.), these dynamic reports help project managers to compare reviewing commonly, find out quality problems, analyze review efficiency, and use review feedback to improve annotation quality.

Payment Dashboard displays the finance of annotation project by different graphical charts. The power of it visualizes the history of annotation work, shows the history of payment to workers, displays the transaction history, calculations of commission, annotation project budget, status of payment and others. Financial dashboard is employed by administrators users to manage the operation expense of various annotation tasks, check the payment to crowdsourcing workers, analyze budget consumption and scope out future cost. Together with annotation analysis tool, financial visualization provides total operation transparency and budget forecasting.

Annotationix provides Project Analytics Dashboards that amalgamates a bunch of visualization techniques into a one page integrated view that provides a ‘global’ picture of a bunch of annotation projects. It displays project completion percentages, dataset utilization percentages, annotation precisions, throughput measures, review metrics, annotation performance measures, crowd worker measures, payout, quality etc. In one report instead of multiple modules. Now, the status of annotation projects can be viewed at one go with one page instead of visiting n number of modules.

Another advantage of the interactive visualization is the usability of the platform is improved. In order to achieve a report, user can easily select or drilled down some certain business data (e. G. Project, Organization, Annotation Category, Dataset Type, User Role, Task Status, Review Stage, Date Range, etc.) and get the visualization reports as they want. It helps the project manager to analyze the operation in different viewpoints without creating a lots of reports manually, and thus improve the efficiency for analysis significantly with the result reports still flexible.

Communication can however be aided by the Annotix visualization. Clients, project managers and higher management are able to get an intuition of the annotation process through these visually represented dashboards. Graphically represented dashboards, as compared to raw database tables or numbers in a report, allows clients and managers a relative understanding of annotation state, employee performance, annotation quality and operation cost. This can prove more useful when reporting to clients.

Thirdly, the visualization capabilities of the system can constitute as part of the decision support. All kinds of graphics shown in management of annotation projects can select problems on workflow, interpretation and modification of work force (distribution statistics and reviewer statistics) and planning and scheduling (project duration and workflow duration statistics) more

efficiently. Visual analytics is no longer a reporting system but a decision support system. As all systems are to be based on the data contained within the central PostgreSQL database the level of consistency, accuracy and up-to-date ‘reporting’ and updating is maintained throughout much of the organization. With a safe and reliable back-end system, 1 source of data and a flexible approach to the dashboards the project statistics are always current and valuable throughout the annotation.

Overall, Annotix visualizations equips a full analytics viewing solution that tracks analytics of annotation, contributor work, payments and operational workflow, and project progress across multiple annotation projects. Annotix brings business operational decision making power in annotation, provides the app to optimize annotation quality, and a full end-to-end visibility in a cloud based Web-platform by using pie, line, bar, summary card graphs, quality dashboards, payment reports, project analytics, and many other set of varied filters.

## 6.2 Univariate and Bivariate Analysis

In order to give relevant information on the annotation projects, this is Annotix is using both Univariate and Bivariate analysis to combine project analytics and reporting. With these analysis the administrators and project managers can visualize the annotation activity, watch over the project efficiency and progress, along as its operating trends. Moreover, in Annotix, the analysis doesn’t refers to the features of the machine learning framework but instead works on operational projects data, taken through whole annotation life cycle, related to datasets, jobs, reviews, worker efficiency and execution flows.

The univariate analysis which is simple descriptive statistics description, it’s would look at each variables separately. It would look at the general overview of the project. They are used to summarizes some general operational stats which include no. Of uploads, dataset categories, annotation job states, project performance, user roles, annotation categories, review states, payment, labor power etc.

The counts of datasets that have been submitted per project, the counts of tasks that are PENDING, IN PROGRESS, SUBMITTED, APPROVED or REJECTED, the counts of users that have been assigned to each system role (Company Administrator, Annotator, Reviewer, System Administrator). The counts of dataset types (image, video, audio or text) and categories (if provided) as well as the counts of annotated images, images and datasets that have been approved or rejected by a reviewer. These frequency counts, percentage results and more detailed analyses can give an administrator a quick reflection of how the project is progressing and at what levels of resources it is using and thus make an informed decision accordingly. Besides other, one more application is worker activity monitoring by using Univariate Analysis. Each of manageability, for example, can measure the sum of annotation by each annotator at the general, review time of each sum of annotation, sum of approved annotation at the general, the rate of permission and the deadline of project for each separately, for project planning, work management, and operation analysis without complex statistical analysis.

Bivariate Analysis-if you are using two different variables/factors for the team or project. This is an extension of the univariate analysis, where the specific variables are not considered but the method they interact and any trends and relationships that it has. An application example of this is when finding the way in which the interaction of the variables affects the quality and time in which the project is completed, aiding company decision-making.

Within Annotix, several important relationships can be analysed using bivariate analysis, including:

The connection between the number of data items (which is plotted on the X- line) and the period that the task was finished (which was on the y-axis (a7))) serves for the administrator to gauge out the amount of time needed for handling a current job to finish.

Characterization of the relationship among project complexity and annotation productivity helping the enterprises to analyse the influence of project requirements on task efficiency

\* To examine the productivity on project basis with the overall quality of annotation to provide the managers that they should not hurry to finish the annotation.

User roles and effects on task rate, giving an indicator of the worker efficiency at a task rate per annotator, reviewer and admin user role.

7) Workload and worker performance; instructors possess the knowledge of a projected volume of annotation work. They could, under conditions, be capable of distributing the volume of annotation work in such a way that, sufficient work will be assigned to the workers.

\* The boxplot of project of review rejection rate with annotation quality works as a lesson on demands of guideline improvement to reach the quality of annotation required.

The comparison of [payment distribution] with [complete annotation task], can achieve labor forces management and economic plan definition very clearly.

With these relationship, expert can know where is the operation bottleneck, trend of productivity, work imbalance, and work load distribution optimization. Bivariate analysis can also use in annotation guideline and reviewer assessment and quality control.

Thus, the integration of the three types would make the three types narrate a wonderful story together in this multi-dimensional collection of annotation projects. In fact, the univariate analysis would provide even more insights in the Bivariate Analyses as it was a perfect quick summary of the existing situation of the project variables; furthermore, Bivariate Analyses on operational features over annotation process of the project would reflect the causal relationships and interdependencies of those variables. Therefore, through integration of such three types of analyses, a project supervisor can gain a bird's eye view of the project to authorise resources scheduling, annotation standard setting, review process framing, decision-making support, annotation quality assurance and data production in a very efficient way for the great bulk of data annotation for further AI or ML application.

## CHAPTER 7

### METHODOLOGY

Methodology of Annotix, it networks the complete path for annotation process by providing a step by step full lifecycle for annotated data processing for achievement of total annotation requirement starting from upload of data set to be annotated to submission of annotation data for validation and payment of annotation work. Annotix methodology emphasizes on automation in creation of annotation task, annotation task assignment, concept, image, text & concept-image work validation, checking of work, monitoring of various tasks, and management of large number of annotation work with quality, scalability, accurate & efficient.

#### ***Step 1: Dataset Upload***

In the first place, company administrator will firstly create an annotation project, then upload one or more dataset files into this created annotation project through web application of Annotix. Supported data file formats contains image, video, audio and text files etc. Company administrator can freely configure various beneficial parameters of annotation project, such as annotation type, task size, the expense per task, review workflow(L1, L2, maybe also L3) etc; During the uploading procedure, the format, the integrity and the meta info of every file will be examine, meanwhile, the duplicated records will be checked, and should be deleted. Only the valid data could be forwarded for annotation process.

#### ***Step 2: Task Generation***

When the sets are correctly checked, the Task Generation Service split automatically each set into small annotation tasks according to the task size specified by the administrator. The task generation service automatically creates each task id and stores it under the specific project in the correct table of the PostgreSQL database with the status Pending.

#### ***Step 3: Task Assignment***

The generated task is shown to more experienced annotator(s). It depends on project configuration and role privilege. If the task is known to the annotator, that annotator is willing to do the input analysis task; then the task can be known as eligible task. Eligible task can be collected from task pool. Once the eligible task is selected, the task is locked to prevent multiple submissions. But if the task does not complete before the time period, then it released to make available again for other annotators.

#### ***Step 4: Annotation***

At this point, the L1 Annotator is doing this annotation according to the annotation guideline and labelling instruction in this project. The annotators need to tag or label this L1 labelled sample according to annotation interface in this system. Progress will be saved periodically when it is accumulated so that the L1 Annotator still can do on it if it can not do on it within the provided time. When annotation is done, L1 Annotator will submit the task so the status will turn to be Submitted from In Progress.

#### ***Step 5: Multi-Level Review***

The submitted annotation will undergo the platform's of tiered QA. The initial QA will be performed by the L2 reviewer and will check for accuracy, task / project requirements, appropriacy

of the level and completeness. If it would not be possible to accept the annotation, the image will be rejected and returned with comments about corrections needed; then the task will be sequentially sent to the annotator for re-annotating / re-submitting to the platform. For the tasks with one more validation step, once the annotation has been accepted by the L2 reviewer, then it will be routed to an L3 Senior reviewer for a final quality validation before it would be accepted and sent to the final dataset.

### ***Step 6: Final Output & Payment***

Once all the necessary review procedures completed, the accepted annotation would then be added to the project final annotated data and be downloaded by the Company Administrator for Artificial Intelligence & machine learning. Meanwhile, the Payment Management Module would automatically revenue there VIEWERS according to the and be transferred by integrated payment channels. The revenue related data would then be saved into the system record.

In conclusion, the six step procedure defined above, allow us to implement a fully secure and automated Cycle Management for annotation process. As the platform Annotix provides, during the entire annotation task, from the data management, automatic job creation, to the intelligent job distribution, multiple quality controls and the auto-payment.

## **7.1 Data Models**

The data model of Annotix The system The system data model The system data model captures how data is conceptualized, stored and managed throughout the annotation lifecycle. Given that several organizations, projects, datasets, annotation tasks, review workflows and payment records are developed in parallel, it is essential to use an appropriate data model to enable that the system has integrity for the data that it uses, that stored data is always formatted equally and that the system performs. Annotix uses a relational data model by means of an PostgreSQL database (PostgreSQL 09), as it saves data in several interconnected entities associated through primary keys and foreign keys. That avoids data duplication, inconsistencies and ensures data integrity and enables speedy data extraction.

Each database element corresponds to a business object clearly specified and implements one function module on the platform. The relationships among the elements permit the system to maintain an entire history trace for each annotation project: from submitting the dataset, to payment after the last revision, etc. Main data elements involved in Annotix are briefly outlined below:

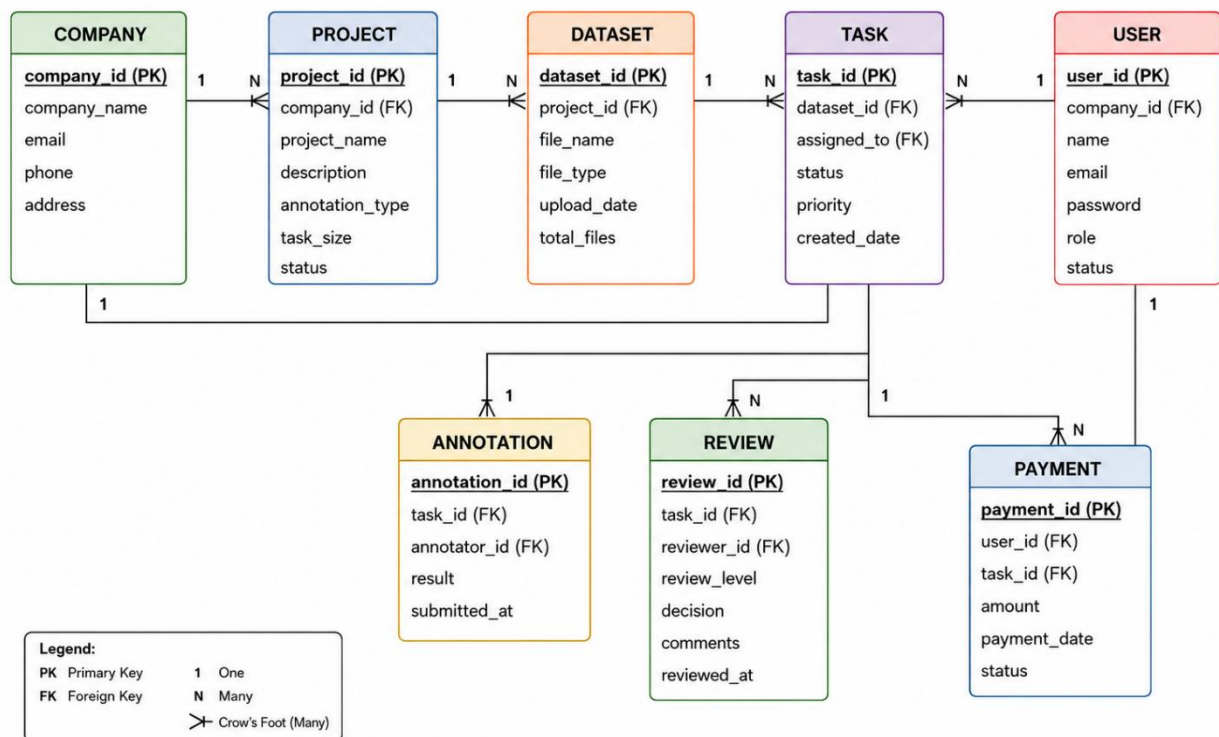
### **Primary data entities:**

- **User** – Saves all the details of every registered user in the platform like login information, user profile information, contact information, user roles like Company Administrator, Annotator, Reviewer, and System Administrator.
- **Company** – The organization for which the platform is being used. The company details are stored in this account, and all projects, users, and datasets within this company reside under this account.
- **Project**- Consists of the configuration of an annotation project that stores project info, annotation type, task size, review workflow, pricing model and project status.
- **Dataset** – Stores the information (metadata) concerning each file uploaded: file description, date of upload, where it is stored, what type of dataset it is, and which project the data is associated with.

- **Task** - an individual annotation task, created automatically on upload of a dataset. Each task has information such as the details of the assignment, the workflow state, priority and current progress during the annotation.
- **Annotation** – Saves the annotation submitted by the given L1 Annotator for the specified task. It saves the annotation, the submission information, time and date of submission, the version history of data etc.
- **Review** – Tracks the quality assurance task conducted by L2/L3 reviewers. Maintains reviewer decision, approval status, rejection status, comments, feedback, and review date/time of the respective annotation.
- **Payment**- Stores the payment data of the annotator and reviewers. Earnings, earnings calculation, commission, history of the Payout, status of the transaction for their annotation work.
- **Audit Log** records all system essential processes in extensive audit logs, including user login, dataset upload, task assignment, annotation download/submission, reviewing/rejection, financial transactions, and system admin's operating activities, etc.

These construct the structure of the Annotix database. By relating all of these the Annotation system is capable of dealing with the user, project, dataset, annotation flow, quality review, payment and the system itself from the Annotation end to end without corrupting or allowing invalid and inconsistent data in the system.

**Figure 7.1: Entity Relationship (ER) Diagram – The Annotix**



**Fig. 7.1: Entity-Relationship (ER) Diagram**

## 7.2 Technology Stack

Annotix follows a three-tier client-server architecture chosen for scalability, ease of maintenance and security, supporting efficient communication between the frontend, backend and database, and concurrent use by many users against a potentially large dataset.

| Component         | Selected Technology | Reason                                       |
|-------------------|---------------------|----------------------------------------------|
| Frontend          | React.js            | Interactive, responsive user interface       |
| Backend           | Spring Boot         | RESTful API development and business logic   |
| Database          | PostgreSQL          | Reliable relational database management      |
| Authentication    | JWT                 | Secure user authentication and authorization |
| Build Tool        | Maven               | Dependency management and project automation |
| API Communication | REST API            | Efficient communication between modules      |
| Version Control   | Git                 | Source code management and collaboration     |

The stack was chosen in order to provide a good compromise between performance and the modularity and flexibility needed for easily adding future features such as AI-assisted annotation and cloud functionality.

## 7.3 Implementation

It is written in the handwritten annotation system was using modular implementation. In it a system has been called as Modular Implementation one, if the system was developed in the way that system was divided into several sub modules, and then those modules were started to develop. Modular implementation was provides key advantage in developing the code for example, debugging was easy, it made the system integration was less complicated and the modules were running successfully at each step.

The implementation of Annotix was divided into the following phases:

- **User login and Role Based Access Control (RBAC):** login with user login on Register Page , login Page and navigates to registration page with user registration, login Page and role management as explained in RBAC for full company admin, annotator, reviewer and system admin.
- **Company & Project Management:** Set up Company Management function, The function of creation of Annotation Project, annotation project management, project configs, management of task size, price management, review management.
- **Dataset Upload and validation** A simple upload of images/video and sound for further processing, extracts data about the file when uploaded, such as the format etc. Stores images/audio/video ready to be used for dataset system. Adds images to collections of datasets with new data, searches for duplicate datasets.

- **Automatic task formation module:** Developed a backend process that takes as input a user uploaded dataset and a user selected project, produces annotation tasks of the dataset according to the project, and saves them into database with correct workflow state.
- **Annotation interface:** for designing a flexible web-based interface which authorisation annotator to claim/comment on the work, annotate on the web, auto\_save the work, submit the work.
- **Multi-level review workflow(L1, L2 and L3).** Developed a three level review process that enabled reviewers to either approve, deny or send back work for editing. Certified on a quality assurance test for the application.
- **Payment management module :** our payment management module will do auto calculation and manage the payment program that would determines every annotator or reviewer's payment based on the completed annotation team.
- **Dashboard and Reporting:** create interactive dashboards and reports on annotation progress, annotation and user performance, annotation and task finished amount, annotation team work efficiency, payment report etc.
- **Scheme of implementation of integration between databases by use of PostgreSQL:** I have designed, tested and integrated the relational database schema to preserve the information's of users, companies, projects, datasets, tasks, annotation, reviews, payments, notifications and audit logs
- **System Testing/Deployment:**8 Performed the functional and integration testing, checked the flows and the security of all the modules and finally deployed the modules to form a fully integrated Cloud based annotation management platform.

The modules are connected together to the other part of the system through secured **RESTful** API by **Spring Boot** framework and all the persistent data are stored in a single **PostgreSQL** database. The deployment of the module in the architecture by component independent, offers scalable, easy-to-maintain and extendable architecture that is room for future evolutions of the application without to drastically modify the system.

Figure 7.2: System Workflow Diagram – The Annotix

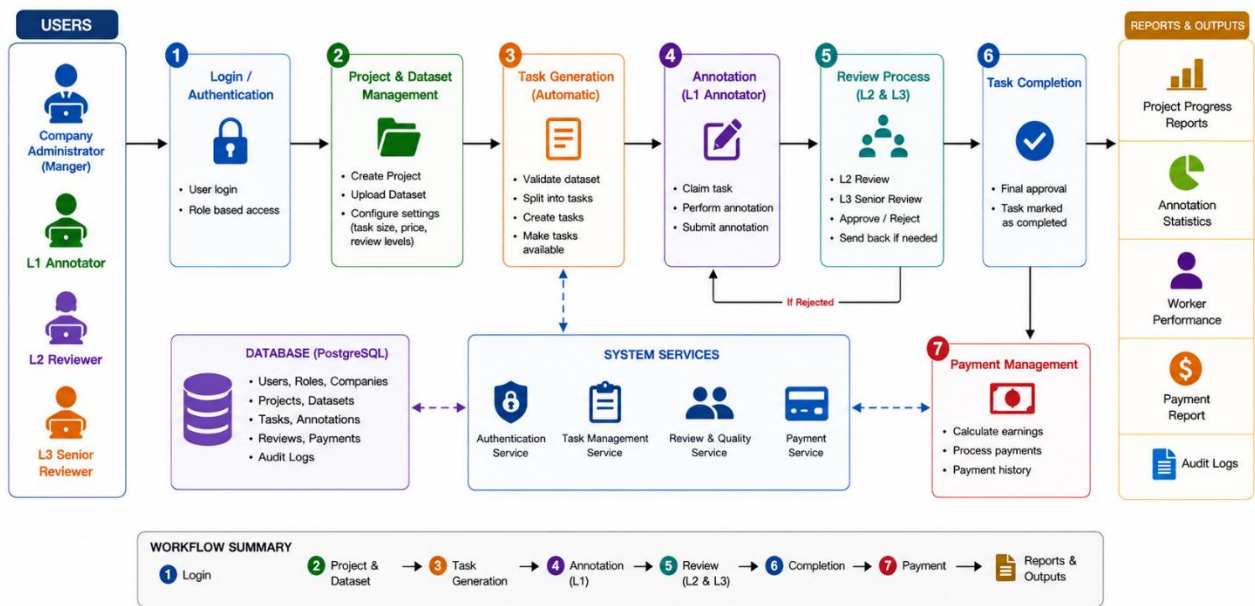


Fig. 7.2: System Workflow Diagram

## 7.4 Results

Annotix was validated on its architectural structure, module design, and business needs to ensure that it meets all the aspects of AI annotation pipeline process. It was tested for each of the user roles, i.e., Company Admin, L1 annotator, L2 reviewer, and L3 reviewer. All its major functionalities such as secure login, project creation, data upload, task creation, annotation, review process, analytics, and payments worked as intended.

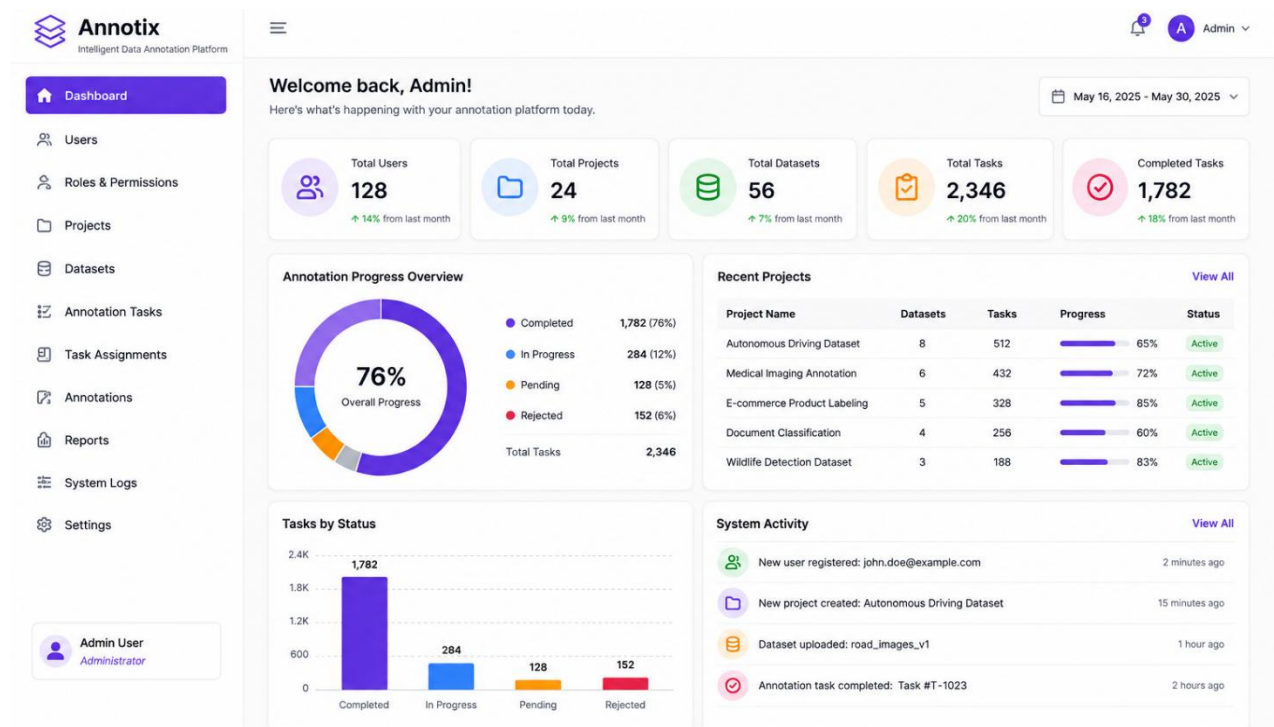


Fig. 7.3: System Dashboard

The user interface that the user sees once he or she logs in is illustrated in Figure 7.3. The interface displays the current tasks of annotation, tasks pending review, total earnings, score, and productivity of the user up to now.

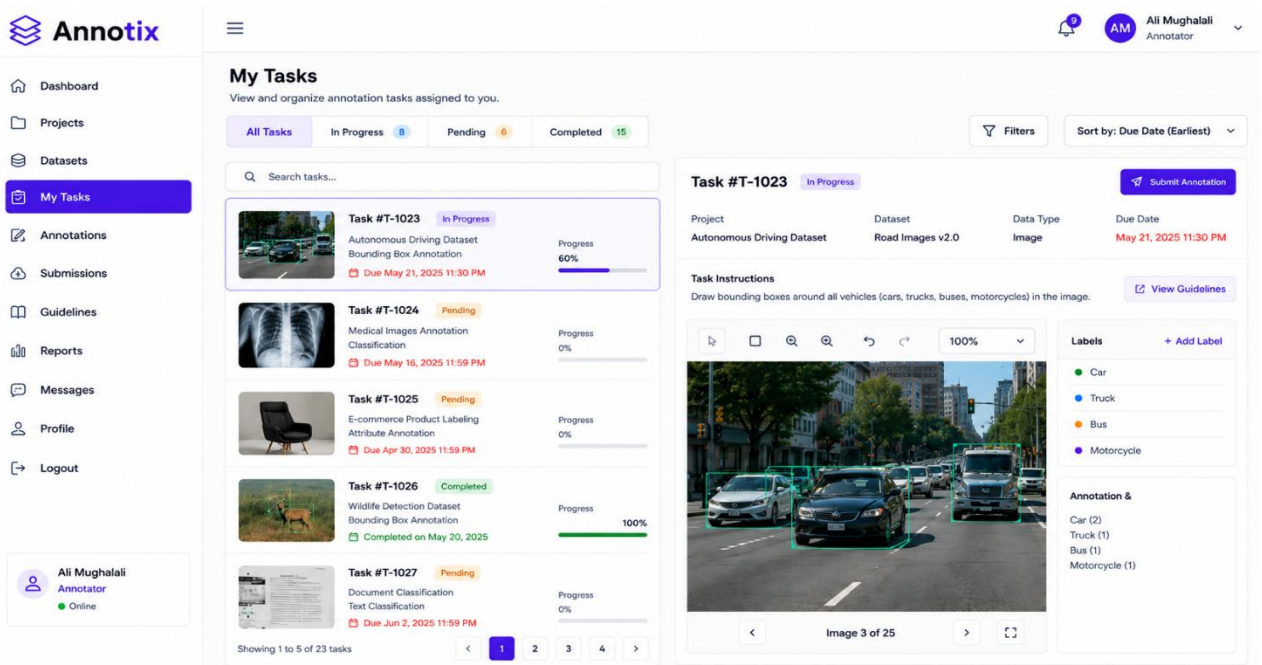


Fig. 7.4: Task Board

Figure 7.4 depicts the Task Board screen, where annotators get to see their tasks and claim the assigned tasks. Task details along with the amount of the reward and the current status of the task are shown on the interface. The task-locking feature makes sure that a particular task is claimed by one annotator at a time.

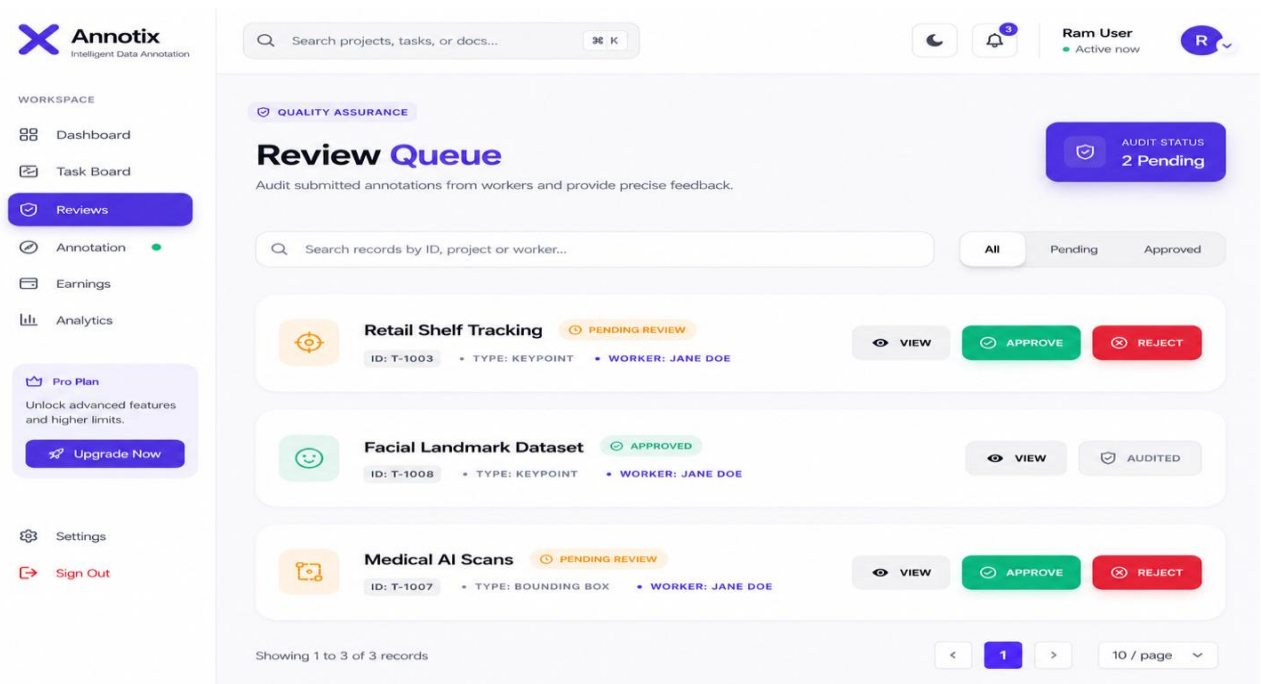


Fig. 7.5: Review Section

As depicted in Figure 7.5 below, the queue for the review of annotation tasks can be accessed by both level two and level three reviewers where submitted annotation tasks can be approved or rejected and comments added before they are either corrected or added to the data set.

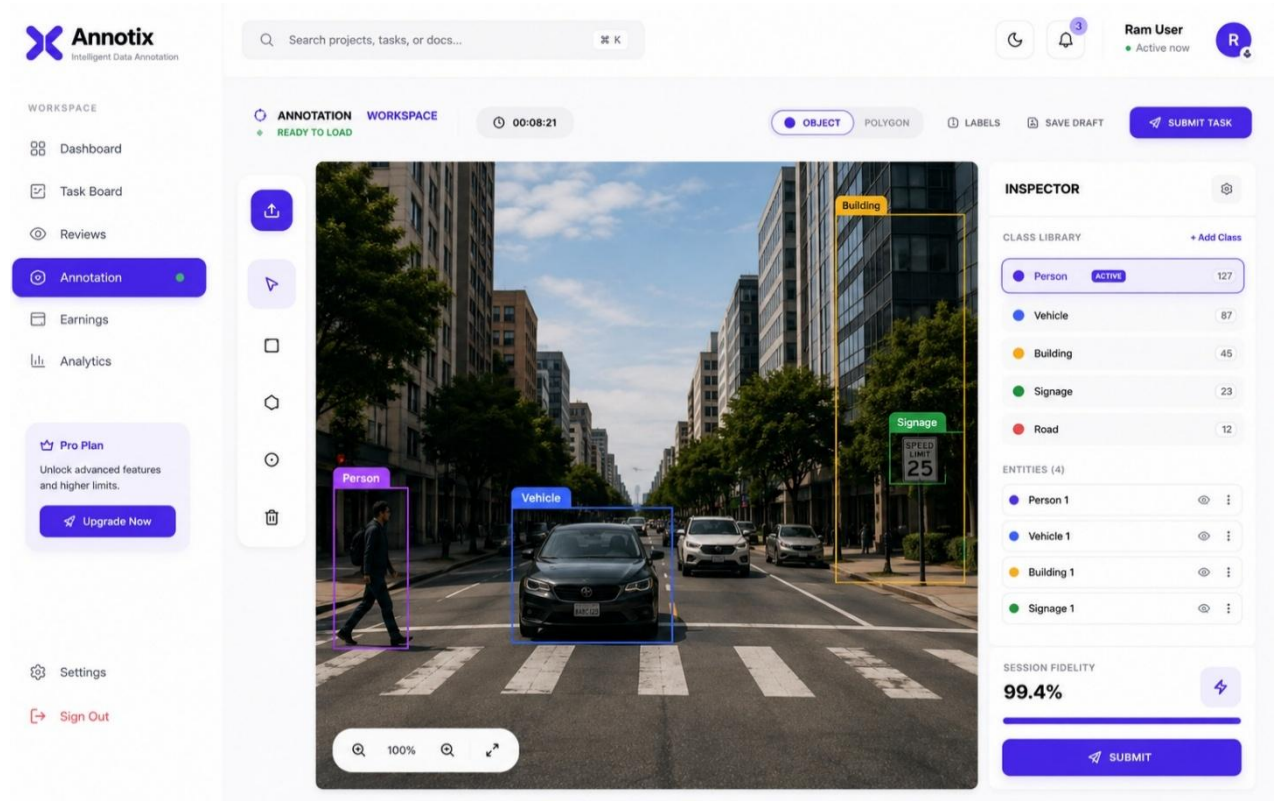


Fig. 7.6: Annotator Workspace

The Annotation Workspace, which is illustrated in Figure 7.6, is the place where the actual process of labelling takes place through the use of annotation tools, tags, and an inspection panel.

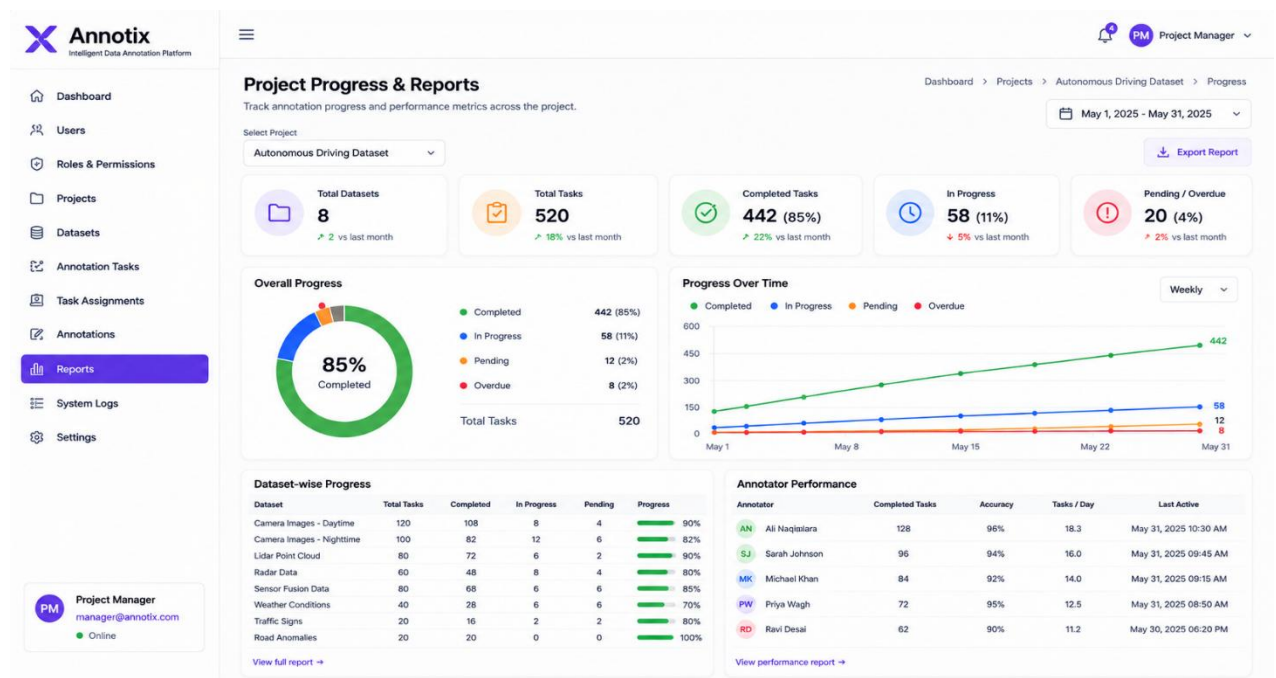


Fig. 7.7: Analytics Hub

Figure 7.7 illustrates the Analytics Dashboard, through which administrators and project managers manage the status of the projects in real time (task was completed, annotation accuracy, worker efficiency and cost), by statistical tables and charts and discover the bottleneck of the project.

The development and testing showed that all the main modules were working smoothly: the login, dataset upload, automatic task generation, annotation, review (approval, rework and rejection), project monitoring, reporting and analytics, payment were all working smoothly, in the smooth flow from the front-end to the back-end and PostgreSQL.

The performance tests demonstrated that the system was responsive with many concurrent users and that the combination of auto task assignment with the locking mechanism prevented the same task from being allocated to two users at once. The combination of JWT authentication and RBAC checks added no noticeable delay to responsiveness.

## CHAPTER 8

### TESTING

Testing. The testing was the most important part, this will make sure the Annotix is working as required and it's stable and secured. All the REST endpoints of the backend have been tested through postman, that enable to check request and response, this will test authentication and validation mechanism directly. Each endpoint has been test with right and wrong input to make sure it produces expected output.

#### 8.1 API Testing Using Postman

Postman was then used to ensure that all the REST end points were accessible and that the requests/responses between frontend and backend were communicating back and forth and that all JWT authorized end points were functioning correctly.

| Test Case ID | API Endpoint          | Method | Expected Result                         | Actual Result                   | Status |
|--------------|-----------------------|--------|-----------------------------------------|---------------------------------|--------|
| TC-01        | /api/auth/login       | POST   | User logs in and receives a JWT token   | Login successful with JWT token | Pass   |
| TC-02        | /api/projects         | POST   | New project created successfully        | Project created                 | Pass   |
| TC-03        | /api/datasets/upload  | POST   | Dataset uploaded successfully           | Dataset stored in database      | Pass   |
| TC-04        | /api/tasks/generate   | POST   | Tasks generated from uploaded dataset   | Tasks generated successfully    | Pass   |
| TC-05        | /api/tasks/{id}/claim | POST   | Annotator claims an available task      | Task assigned successfully      | Pass   |
| TC-06        | /api/annotations      | POST   | Annotation submitted successfully       | Annotation saved                | Pass   |
| TC-07        | /api/reviews/{id}     | PUT    | Reviewer approves or rejects annotation | Review updated successfully     | Pass   |
| TC-08        | /api/payments         | GET    | Retrieve payment details                | Payment data returned           | Pass   |
| TC-09        | /api/dashboard        | GET    | Display dashboard statistics            | Statistics displayed correctly  | Pass   |
| TC-10        | /api/users            | GET    | Retrieve registered users               | User list returned              | Pass   |

The screenshot displays the 'Test Case 1 – User Login API' configuration. The request is a POST to `{{base_url}}/api/auth/login` with a JSON body containing email and password. The response is a 200 OK status with a JSON body containing a success message, a token, and user details.

**Request:**

```
POST {{base_url}}/api/auth/login
```

**Body:**

```
{
  "email": "admin@example.com",
  "password": "Admin@123"
}
```

**Response:** 200 OK · 164 ms · 512 B

**JSON Response:**

```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIjOiEsImVtYWlsIjoiYWRtaW5AZXhhbXBsZS5jb20iLCJy",
    "user": {
      "user_id": 1,
      "name": "Admin User",
      "email": "admin@example.com",
      "role": "ADMIN"
    }
  }
}
```

**Parameters Table:**

| Parameter       | Value               |
|-----------------|---------------------|
| API             | /api/auth/login     |
| Method          | POST                |
| Request Body    | Email, Password     |
| Expected Status | 200 OK              |
| Result          | JWT Token Generated |
| Status          | Pass                |

Fig. 8.1: Login API Response

The screenshot displays the 'Test Case 2 – Create Project API' configuration. The request is a POST to `{{base_url}}/api/projects` with a JSON body containing project details. The response is a 201 Created status with a JSON body containing a success message, a project ID, and project details.

**Request:**

```
POST {{base_url}}/api/projects
```

**Body:**

```
{
  "name": "AI Image Annotation Project",
  "description": "Project for annotating and labeling images for ML model training.",
  "annotation_type": "Image",
  "task_size": 10,
  "review_levels": 2,
  "priority": "High",
  "status": "Active"
}
```

**Response:** 201 Created · 182 ms · 620 B

**JSON Response:**

```
{
  "success": true,
  "message": "Project created successfully",
  "data": {
    "project_id": 101,
    "name": "AI Image Annotation Project",
    "description": "Project for annotating and labeling images for ML model training.",
    "annotation_type": "Image",
    "task_size": 10,
    "review_levels": 2,
    "status": "Active",
    "created_at": "2025-05-24T10:30:45.123Z"
  }
}
```

**Parameters Table:**

| Parameter       | Value                        |
|-----------------|------------------------------|
| API             | /api/projects                |
| Method          | POST                         |
| Authorization   | Bearer JWT Token             |
| Expected Status | 201 Created                  |
| Result          | Project Created Successfully |
| Status          | Pass                         |

Fig. 8.2: Project Creation API

**Test Case 3 – Upload Dataset API**

POST `{{base_url}}/api/datasets/upload` **Send**

Params Authorization Headers (11) **Body** Pre-request Script Tests Settings

☐ none ☒ form-data ☐ x-www-form-urlencoded ☐ raw ☐ binary ☐ GraphQL

| Key                                              | Value                               | Description                           |
|--------------------------------------------------|-------------------------------------|---------------------------------------|
| <input checked="" type="checkbox"/> file         | sample_images.zip                   | Dataset file (Image/Video/Audio/Text) |
| <input checked="" type="checkbox"/> project_id   | 101                                 | ID of the project                     |
| <input checked="" type="checkbox"/> dataset_name | Sample Image Dataset                | Name of the dataset                   |
| <input checked="" type="checkbox"/> dataset_type | IMAGE                               | Type of dataset                       |
| <input checked="" type="checkbox"/> description  | Sample dataset for image annotation | Short description of dataset          |

**Body** Cookies Headers (12) Test Results **200 OK** · 158 ms · 826 B · Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "success": true,
3   "message": "Dataset uploaded successfully",
4   "data": {
5     "dataset_id": 203,
6     "project_id": 101,
7     "dataset_name": "Sample Image Dataset",
8     "dataset_type": "IMAGE",
9     "file_name": "sample_images.zip",
10    "file_size": 5242880,
11    "total_files": 1250,
12    "status": "UPLOADED",
13    "uploaded_by": 1,
14    "uploaded_at": "2025-05-24T10:45:32.123Z"
15  }
16 }
```

| Parameter       | Value                         |
|-----------------|-------------------------------|
| API             | /api/datasets/upload          |
| Method          | POST                          |
| Request Type    | Multipart Form Data           |
| Expected Status | 200 OK                        |
| Result          | Dataset Uploaded Successfully |
| Status          | Pass                          |

Fig. 8.3: Dataset Upload API Response

**Test Case 4 – Generate Tasks API**

POST `{{base_url}}/api/tasks/generate` **Send**

Params Authorization Headers (10) **Body** Pre-request Script Tests Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON**

```
1 {
2   "project_id": 101,
3   "dataset_id": 203,
4   "task_size": 10,
5   "priority": "High",
6   "auto_assign": true
7 }
```

**Body** Cookies Headers (12) Test Results **200 OK** · 156 ms · 710 B · Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "success": true,
3   "message": "Tasks generated successfully",
4   "data": {
5     "project_id": 101,
6     "dataset_id": 203,
7     "task_size": 10,
8     "total_tasks": 1250,
9     "tasks_created": 1250,
10    "status": "GENERATED"
11  }
12 }
```

| Parameter       | Value                        |
|-----------------|------------------------------|
| API             | /api/tasks/generate          |
| Method          | POST                         |
| Expected Status | 200 OK                       |
| Result          | Tasks Generated Successfully |
| Status          | Pass                         |

Fig. 8.4: Task Generation API Response

HTTP Test Case 5 – Submit Annotation API

Save

POST {{base\_url}}/api/annotations Send

Params Authorization Headers (11) Body Pre-request Script Tests Settings Cookies Beautify

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL ☒ JSON

```
1 {
2   "task_id": 501,
3   "annotator_id": 201,
4   "annotation_data": {
5     "image_id": "img_0001.jpg",
6     "objects": [
7       { "label": "car", "bbox": [120, 150, 320, 280] },
8       { "label": "person", "bbox": [400, 200, 480, 450] }
9     ]
10  },
11 }
12 }
```

Body Cookies Headers (12) Test Results 201 Created · 168 ms · 612 B · Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "success": true,
3   "message": "Annotation submitted successfully",
4   "data": {
5     "annotation_id": 7891,
6     "task_id": 501,
7     "annotator_id": 201,
8     "submitted_at": "2025-05-24T11:20:45.123Z",
9     "status": "SUBMITTED"
10  }
11 }
12 }
```

| Parameter       | Value                             |
|-----------------|-----------------------------------|
| API             | /api/annotations                  |
| Method          | POST                              |
| Expected Status | 201 Created                       |
| Result          | Annotation Submitted Successfully |
| Status          | Pass                              |

**Fig. 8.5: Annotation Submission API Response**

## CHAPTER 9

### SDG MAPPING

This is one of the key objectives of the United Nations Sustainable Development Goals (SDGs): achieving sustainable growth by driving technological innovation and protecting the environment. Modern software systems must address business issues and deliver societal value through productivity improvements, employment and the responsible use of innovation.

The Annotix platform intends to be a durable tool, contributing to sustainable development, by providing a single cloud annotation management platform allowing to facilitate and automate the acquisition of high quality datasets for Artificial intelligence (AI)&Machine learning(ML). The solution from Annotix implements several steps of projects including dataset storage & automation, tasks distribution, annotation, validation, payment and project monitoring. The company aims at easing the shift to digital technologies, within a collaborative ecosystem allowing the construction of strong Artificial Intelligence, while providing good working conditions to annotation and review participants.

#### 9.1 Selected SDG Goal(s)

The chance related to the Annotix solution is mainly focusing on SDG 9 on “Industry, Innovation and Infrastructure”. Indeed, applications of AI are demanding annotation of very big datasets to train good models. Annotix can contribute to face this problem because it proposes a highly automated cloud-based platform that accelerates the annotation workflows and improve dataset quality through a review step.

It also addresses the Sustainable Development Goal 8- Decent Work and Economic Growth, because it provide structured digital jobs to the task performers (annotators, reviewers and quality analysts). Its platform enables firms to leverage digital distributed workforces, distributing the tasks, managing the payments and protecting the projects from the counterparty.

| SDG Goal                                                                                     | Title                                   | Contribution of Annotix                                                                                      |
|----------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <br>SDG 9 | Industry, Innovation and Infrastructure | Develops a modern cloud-based annotation platform that supports AI innovation and digital infrastructure.    |
| <br>SDG 8 | Decent Work and Economic Growth         | Creates structured employment opportunities through transparent annotation workflows and payment management. |

#### 9.2 Specific Targets Addressed

| SDG Goal | Target | Description                                               | Contribution                                                          |
|----------|--------|-----------------------------------------------------------|-----------------------------------------------------------------------|
| SDG 9    | 9.5    | Enhance scientific research and technological innovation. | Supports AI development by producing high-quality annotated datasets. |

| SDG Goal | Target | Description                                                   | Contribution                                                                        |
|----------|--------|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
| SDG 9    | 9.c    | Increase access to information and communication technology.  | Provides a cloud-based annotation platform accessible through the web.              |
| SDG 8    | 8.2    | Achieve higher productivity through technological innovation. | Automates the annotation workflow and reduces manual administrative work.           |
| SDG 8    | 8.5    | Achieve productive employment and decent work.                | Provides transparent task allocation and payment management for annotation workers. |

These objectives are directly given support from the architecture of the platform, as the merging of the latest practices in software engineering and cloud computing is used to optimize the annotation process, management of the workforce, and digital cooperation.

### 9.3 Social Impact

Annotix not just aims to speed up data annotation work and management but to also affect an organization, designers and the entire AI ecosystem positively. Therefore, since AI and ML are quickly expanding to many domains, such as health, education, agriculture, self-driving cars, finance, manufacture and Internet of Things and E-Commerce, the demand for a mass amount of annotated data will subsequently raise. The execution of data annotation with the secure, scalable, and cooperative annotation platform Annotix, aims to fulfil such needs and establish a meaningful digital labor market for organizations.

**Digital job:** The main contribution the system made was enabling the digital job. Annotix, through its working environment that include task assignment, working flow, task progress and automatic payment computing, support and guide its Annotator, Reviewer, and all kind of Admins during their work. Because users can participate in annotation based on their role, so that any students, freelancer, research staff, professionals who looking for flexibility and freelance from distant can participate in Annotix projects. Tasks, reviewers and payment computing will be totally transparent.

In addition, it supports long distance collaboration and digital inclusion. Being a web application hosted on cloud, any user can login to the system easily at any place and at any time as long as he or she has an internet connection. With this, firms/organisations could create teams of annotators all over the world irrespective of physical and geographical distance, even without the need for physical office buildings. So people of remote or under-developed areas (e.g. small cities, impoverished communities, etc.) can use the cloud based system to contribute to improving AI and can get employment opportunities for themselves.

A second added social value of Annotix comes in the improvement of annotation quality and thus in higher data reliability. The more qualitative the annotation, the higher the accuracy and reliability of AI systems. The use of an automatic workflow, sub-tasks division and three levels of review (L1, L2, L3) and discussion provided by Annotix help to improve annotation quality and reduce the inter-annotator variability before the final dataset is created. The more qualitative

annotation is relevant for high-reliable AI solutions to be used for medicine, self-driving cars, language technology, fraud detection, smart agriculture, customer support or educational systems.

One more huge advantage of Annotix is its capability of making a crowdsourced annotation task transparent and accountable. All critical activities like data uploads, task assignment, annotation deliveries, reviewers' decisions, payments & administrative processes are recorded in Annotix. Through interactive reporting capabilities and audit trails, the administrators can constantly track the progress of a project, monitor the worker performance, check annotation accuracy, and analyse performance bottlenecks in the annotation pipeline.

From an educational point of view Annotix is an interesting environment where both students, researchers and software developers can work on. A student working with Annotix will have experience on how is annotation work done on real environment, how is a modern annotation project run, how can an cloud application be created, how the quality should be checked, how can software projects be organized in a system etc. Researchers and universities can make good quality data sets useful for research purposes and meanwhile offer a valuable student experience on the current technology state for AI.

Automate more administrative tasks: the platform can automate routine manpower administration and management work, such as auto-creation of activities and tasks, equitable and intelligent distribution of tasks, centralizing the entire management, automated bill and payment monitoring, automatic generation of final project report, etc. As a result, not only the overall operating costs can be greatly reduced, but also project leaders can focus on ensuring project quality while saving themselves the time of repetitive tasks.

Furthermore, Annotix has also contribution on fostering the AI ecosystem, by supplying reliable, accurately labelled, extremely coherent and structured source of label data for use in training using supervised ML approaches. With the increasing demand to construct trust-worthy AI systems, quality training data source as provided by Annotix has becomes even more important to support scalable, transparent and standard-based annotation process.

In conclusion, Annotix is committed to all kinds of social values, like a digitalized job provision; decentralized virtual working environment which makes people get together to contribute on the annotation task; open process of working; quality-driven annotation pipeline; learning centre based on artificial intelligence, and expanding enterprise automation service etc. It's not only helping accelerate workflow of the data annotation process, but also delivering to the cloud-based service without individual people to invest more on assisting the humans.

## **9.4 Environmental Sustainability**

Annotix enables paperless annotation management: projects are configured, tracked and annotated completely on screen replacing all that paper based record-keeping, sign-offs and physical files associated with most projects, all datasets, tasks, review records and payment history are managed in the cloud.

Given that the platform is designed to run on a browser, the project administrator, annotators, reviewers and the project manager can be based in different places, free of having to commute in a shared office, possibly lowering the environmental impact in terms of emissions.

Crucially, however, such automated processing is also more efficient with IT resources. Automated task generation, digital tracking of progress and integrated rework management minimise idle processing time and optimise the use of server and disk space through a well designed relational data model built on PostgreSQL.

Decreasing dependency on tangible infrastructure makes for one more improvement in sustainability; organisations are able to administer the annotation activity, monitor results and create reports on the platform without generating paper or needing the printers, paper, toner and other associated expenses.

The cloud deployment option employed by annotix also enables hardware to be utilised more efficiently as computing and storage requirements can grow in line with projects instead of organisations having to buy new physical hardware beforehand thus also reducing the electronic waste created by frequent hardware upgrades.

An electronic system can by default ease the print/circulation/preparation of status reports, which are automatically generated and instantly IT-enabled.

The modular design of the platform extends the value of its useful life: new features can be added to the existing system as a layer rather than force a complete redevelopment of the annotation platform for an organisation, thus reducing the time and resource cost of platform development and evolution.

The aforementioned design features of Annotix (less paper, facilitating remote working, optimization of resources, and cloud deployment with a low footprint) collectively contribute to a fairly modest but decidedly beneficial sustainability profile in comparison with conventional annotation workflows, which rely heavily upon paper and office space.

## 9.5 Innovation Relevance

Annotix integrates the core requirements of Web 2.0, Cloud Computing, automating of the annotation workflow and securing data, all in a single management system. As there is an increased use of AI, NLP and ML the volume of accurately annotated data that will be required is only going to increase, the cost of managing that workflow across multiple tools, tabs and emails becomes prohibitive. Annotix solves that through

A novel feature of the platform is the automatic creation and assignment of annotation tasks. The image data is uploaded and the annotation task is created and assigned to the annotators automatically, and manual monotonous operations are minimized.

Another distinction made is within the multiple level quality review system. Instead of relying on a single quality review as most annotation tools do, the Annotix takes advantage of three different stages of review: L1 Annotator, L2 Reviewer, and L3 Super Reviewer; thereby, building up the AI/ML model step-by-step, with a significant boost in output data quality.

The platform has security built in, through the use of RBAC and JWT based authentication the user has access to only that data and that functionality that they are allowed. This will allow different organisations to share the same platform, in their own private instance.

The Annotix application is developed in React.js, Spring Boot and PostgreSQL by a modular architecture for scalability reasons: The React.js frontend provides a fast and friendly UI; The Spring Boot backend provides an easy to integrate set of REST APIs that the components can rely on; and finally PostgreSQL is providing a flexible database for hosting and quickly querying the datasets, projects, user, annotation and payment information.

Beyond the quick-click annotation and review Annotix provides live project monitoring and analytics dashboards for the administrator. The administrator can keep an eye on workflow progress, annotation completion rates, worker productivity and reporting in one screen and make quick data-based decisions for work force distribution and promptly detect bottlenecks.

Integrated management of payment is the second advantage since most annotation system doesn't maintain the payment inside the system; admin can review the work and also control the payment on the work in system directly.

Because it is a modular system, Annotix can tolerate the integration of future enhancements without significantly changing the existing structure, such as an ML guided quality assurance layer, in-browser collaborative annotation, ML generated annotation suggestions and further cloud storage integration.

## CHAPTER 10

### CONCLUSION

Annotix was conceived, built, deployed and is now being used as a complete cloud based annotation management platform trying to resolve the unavoidable drawbacks and issues in ad-hoc, manual workflow during the process of annotation. We wanted to provide an end to end platform along with dataset management, annotation management, task delegation, multi-level quality control, payment mechanisms, and project management; the system was a success in achieving this.

Streamlines annotation workflow by eliminating the old hodgepodge of manual task assignment and disparate project management tools with a single, automated workflow. Datasets can be uploaded, annotation tasks created, and then tasks automatically generated and managed by a live, streaming progress dashboard. Annotations can go from assignment to submission, and to a checkpoint at the right quality, prior to final data set assembly.

However, by using React.js at frontend, Spring Boot at backend, PostgreSQL at the data processing layer we create a good, scalable architecture which can support a lot of users at the same time. The addition of JWT authentication and the use of RBAC framework ensure that users have a secure experience and that project data is not available outside of users own role, and the application is built with modularity in mind, which should support it well.

A test run was conducted to ensure that the minimum features of the platform were working. The minimal features include user and project management and maintaining, dataset uploading, task creation and submission, annotation, status update of the payments, and reporting by dashboard. The API tests were performed by the use of Postman tool to check the end points behave as required. System test were performed to test the accuracy and reliability of the system.

Besides its technical goals, Annotix has the following high-level purposes related to the Digital Transformation: increase productivity, reduce administration costs and enable remote collaboration. It also supports the UN SDGs through the support to SDG 9 (Industry, Innovation and Infrastructure) by the solid infrastructure it has and through SDG 8 (Decent Work and Economic Growth) by decent working conditions for annotations and reviews.

This project demonstrated that a unified platform to handle the variety of needs will greatly facilitate the efficiency, quality, transparency, and scale of annotation tasks and establish a firm basis for more advanced augmentations driven by AI, such as ML-assisted annotation, automatic QA, and deeper analytics.

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## CHAPTER 11

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## CHAPTER 12

### APPENDIX

#### Source Code

Below is a few of the application backend elements from the Annotix system, showing the entity, the repository, the service, the controller and the security levels. These are only parts of the source code.

##### A.1 User Entity

*File: User.java    Language: Java (Spring Boot / JPA)*

Defines the User entity persisted to the users table, including the role assigned to each account (Company Admin, L1 Annotator, L2 Reviewer, L3 Super Reviewer or System Admin).

```
package com.annotix.user.entity;

import jakarta.persistence.*;
import lombok.*;

@Entity
@Table(name = "users")
@Getter @Setter
@NoArgsConstructor @AllArgsConstructor @Builder
public class User {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false)
    private String fullName;

    @Column(nullable = false, unique = true)
    private String email;

    @Column(nullable = false)
    private String password;

    @Enumerated(EnumType.STRING)
    private Role role;

    @Column(nullable = false)
    private boolean active = true;
}
```

##### A.2 Project Entity

*File: Project.java    Language: Java (Spring Boot / JPA)*

Defines the Project entity: contains the configuration set by the company administrator during creation of a new annotation project (annotation type, number of files/tasks)

```
package com.annotix.project.entity;

import jakarta.persistence.*;
```

```

import lombok.*;
import java.time.LocalDateTime;

@Entity
@Table(name = "projects")
@Getter @Setter
@NoArgsConstructor @AllArgsConstructor @Builder
public class Project {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long projectId;

    private String projectName;
    private String annotationType;
    private Integer totalFiles;
    private Integer taskSize;
    private LocalDateTime createdAt;
    private String status;
}

```

### A.3 Task Repository

*File: TaskRepository.java Language: Java (Spring Data JPA)*

Repository interface providing access to lookups utilized by the task workflow searching for tasks by status (for the availabletask pool) and by assigned user(for an annotator's task list).

```

package com.annotix.task.repository;

import com.annotix.task.entity.Task;
import com.annotix.task.entity.TaskStatus;
import org.springframework.data.jpa.repository.JpaRepository;
import java.util.List;

public interface TaskRepository extends JpaRepository<Task, Long> {
    List<Task> findByStatus(TaskStatus status);
    List<Task> findByAssignedUserId(Long userId);
}

```

### A.4 Task Service — Claim Logic

*File: TaskService.java Language: Java (Spring Boot)*

This enforces the task-locking rule from section 4.3; only an AVAILABLE task can be locked and only an available transaction can claim a locked task, as well as changing its status to IN\_PROGRESS.

```

package com.annotix.task.service;

import com.annotix.task.entity.Task;
import com.annotix.task.entity.TaskStatus;
import com.annotix.task.repository.TaskRepository;
import lombok.RequiredArgsConstructor;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
@RequiredArgsConstructor

```

```

public class TaskService {

    private final TaskRepository repository;

    @Transactional
    public Task claimTask(Long taskId, Long userId) {
        Task task = repository.findById(taskId)
            .orElseThrow(() -> new RuntimeException("Task not found"));

        if (task.getStatus() != TaskStatus.AVAILABLE) {
            throw new RuntimeException("Task already claimed");
        }

        task.setAssignedUserId(userId);
        task.setStatus(TaskStatus.IN_PROGRESS);
        return repository.save(task);
    }
}

```

### A.5 Task REST Controller

*File: TaskController.java Language: Java (Spring Boot)*

Exposes the claim-task endpoint referenced in the API test cases (Chapter 8, TC-05) at POST /api/tasks/{id}/claim.

```

package com.annotix.task.controller;

import com.annotix.task.entity.Task;
import com.annotix.task.service.TaskService;
import lombok.RequiredArgsConstructor;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/api/tasks")
@RequiredArgsConstructor
public class TaskController {

    private final TaskService taskService;

    @PostMapping("/{id}/claim")
    public ResponseEntity<Task> claimTask(
        @PathVariable Long id,
        @RequestParam Long userId) {
        return ResponseEntity.ok(taskService.claimTask(id, userId));
    }
}

```

### A.6 JWT Authentication Filter

*File: JwtAuthenticationFilter.java Language: Java (Spring Security)*

A servlet filter to parse out and validate the JWT every time an incoming request comes through, also storing the security context so downstream controllers can perform role based permissions.

```

package com.annotix.security;

import jakarta.servlet.*;
import jakarta.servlet.http.*;

```

```

import lombok.RequiredArgsConstructor;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.stereotype.Component;
import java.io.IOException;

@Component
@RequiredArgsConstructor
public class JwtAuthenticationFilter extends GenericFilter {

    private final JwtService jwtService;

    @Override
    public void doFilter(ServletRequest request, ServletResponse response,
        FilterChain chain) throws IOException, ServletException {
        HttpServletRequest http = (HttpServletRequest) request;
        String token = jwtService.extractToken(http);

        if (token != null && jwtService.isValid(token)) {
            SecurityContextHolder.getContext()
                .setAuthentication(jwtService.getAuthentication(token));
        }
        chain.doFilter(request, response);
    }
}

```

### A.7 Review Service — Approve / Reject Logic

*File: ReviewService.java Language: Java (Spring Boot)*

For the multi level review decision as described in 3.3 and 4.2 implemented. Records reviewer decision and comments, and moves the task to APPROVED / REWORK.

```

package com.annotix.review.service;

import com.annotix.review.entity.Review;
import com.annotix.review.entity.ReviewDecision;
import com.annotix.review.repository.ReviewRepository;
import com.annotix.task.entity.Task;
import com.annotix.task.entity.TaskStatus;
import com.annotix.task.repository.TaskRepository;
import lombok.RequiredArgsConstructor;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
@RequiredArgsConstructor
public class ReviewService {

    private final ReviewRepository reviewRepository;
    private final TaskRepository taskRepository;

    @Transactional
    public Task submitReview(Long taskId, Long reviewerId,
        ReviewDecision decision, String comments) {
        Task task = taskRepository.findById(taskId)
            .orElseThrow(() -> new RuntimeException("Task not found"));

        Review review = Review.builder()
            .taskId(taskId)
            .reviewerId(reviewerId)

```

```

        .decision(decision)
        .comments(comments)
        .build();
reviewRepository.save(review);

task.setStatus(decision == ReviewDecision.APPROVED
    ? TaskStatus.APPROVED
    : TaskStatus.REWORK);
return taskRepository.save(task);
    }
}

```

## A.8 Database Schema Extract

*File: schema.sql Language: SQL (PostgreSQL DDL)*

The simplified data definition language for the annotation `tasks` and `reviews` tables from throughout Chapters 4 and 7, showing all of the foreign-key relationships to `projects` and `users`, and the fields used by the locking and review workflow.

```

-- annotation_tasks table (simplified)
CREATE TABLE annotation_tasks (
    id          BIGSERIAL PRIMARY KEY,
    project_id  BIGINT REFERENCES projects(project_id),
    assigned_user_id BIGINT REFERENCES users(id),
    status      VARCHAR(20) NOT NULL DEFAULT 'PENDING',
    locked_at   TIMESTAMP,
    lock_expires_at TIMESTAMP,
    created_at  TIMESTAMP NOT NULL DEFAULT now()
);

-- reviews table
CREATE TABLE reviews (
    id          BIGSERIAL PRIMARY KEY,
    task_id     BIGINT REFERENCES annotation_tasks(id),
    reviewer_id BIGINT REFERENCES users(id),
    decision    VARCHAR(20) NOT NULL,
    comments    TEXT,
    reviewed_at TIMESTAMP NOT NULL DEFAULT now()
);

```

# Lokesh Ck

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